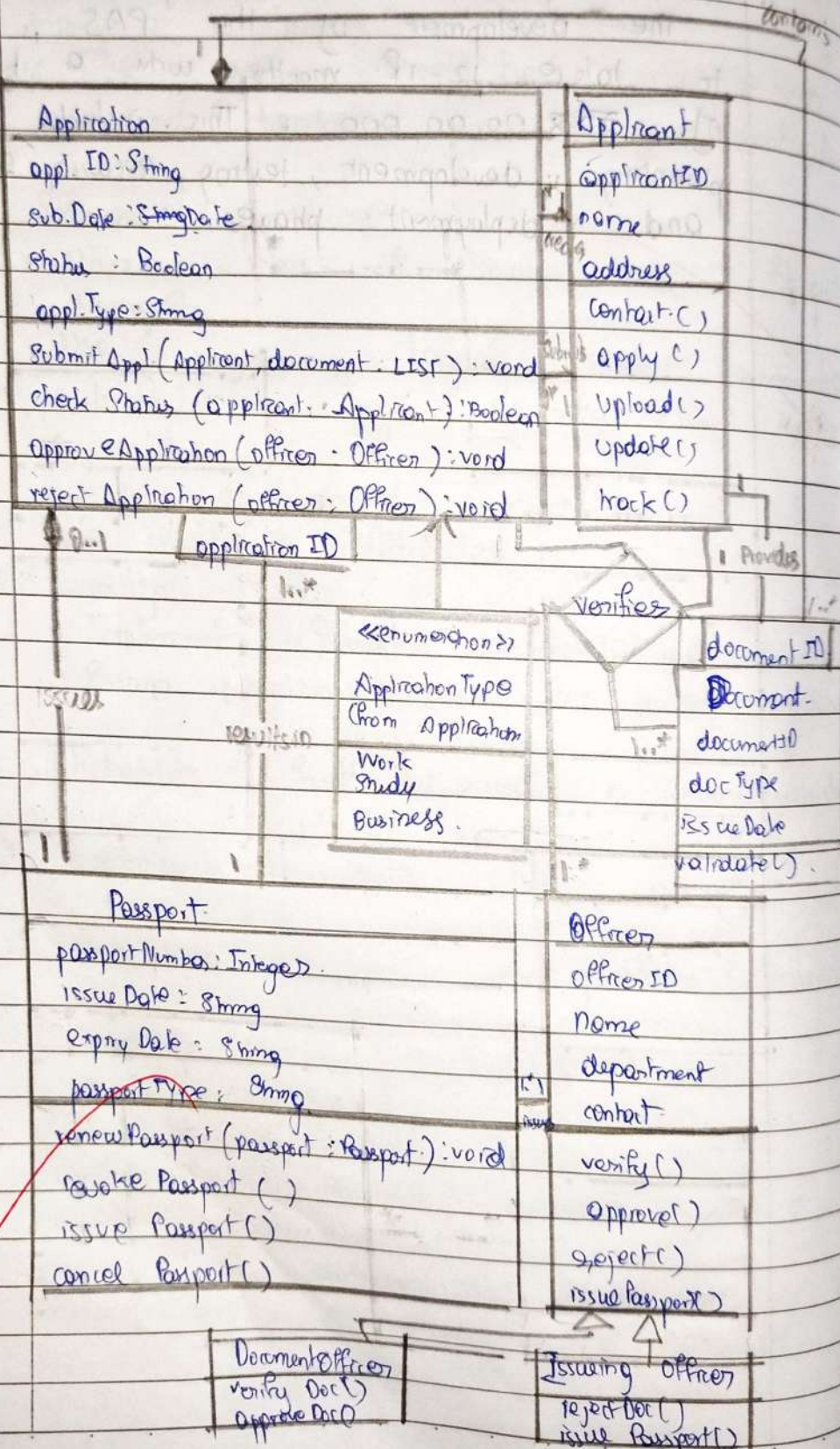


① PASSPORT AUTOMATION SYSTEM



HOTEL MANAGEMENT SYSTEMS

Hotel

hotelid : int
name : string
address : string
phone : string
email : string
addroom()
get-available-rooms()

Room

roomid : int
roomnumber : string
type : string
status : string
price : double
check-availability()
updatestatus()

staff

id : int
name : string
phone : string
role : string
performduty()

Reservation

id : int
checkin date : Date
checkout : Date
status : Success
confirm reservation()
cancel reservation()

Receptionist

specialization : string
checkin guest()
checkout guest()

Manager

department : string
general report()

Service

id : int
Type : string
cost : double
assign to guest()

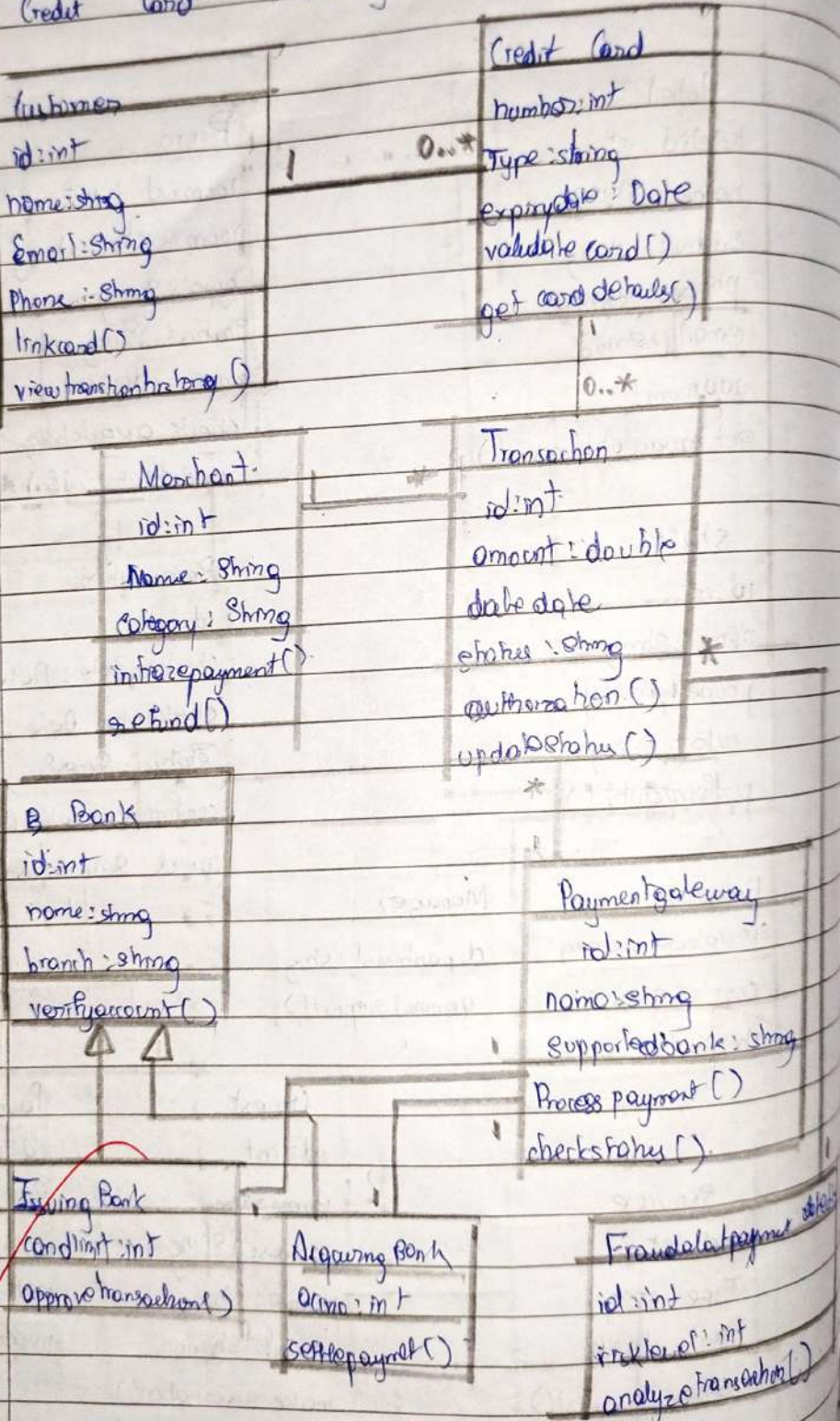
Guest

id : int
name : string
address : string
phone : string
email : string
make reservation()

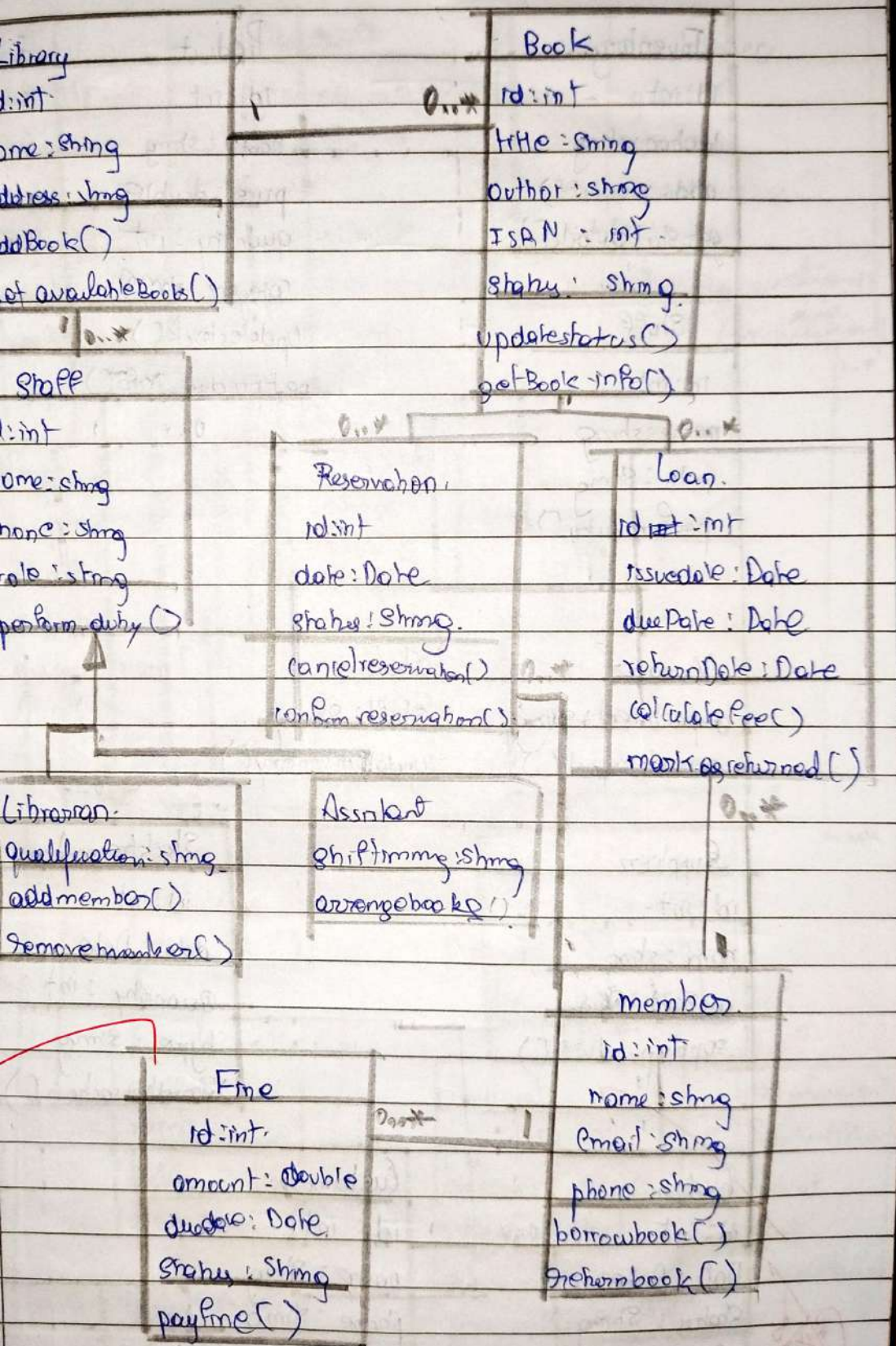
Payment

id : int
Amount : double
date : Date
method : string
processing payment()

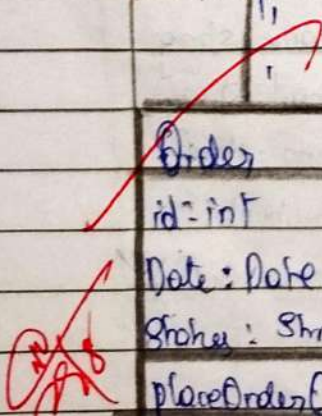
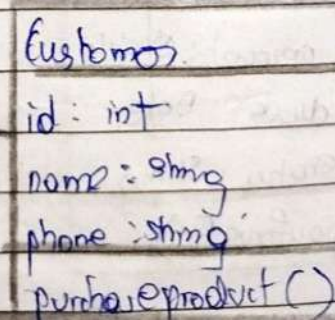
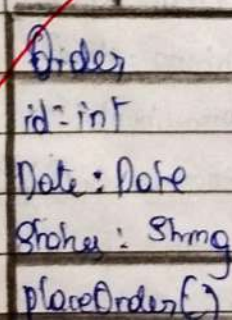
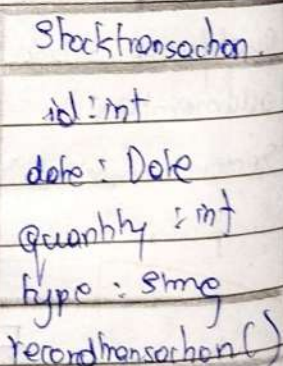
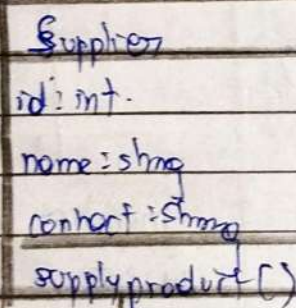
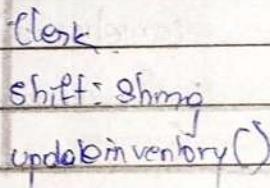
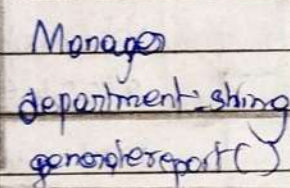
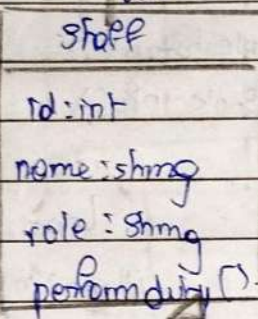
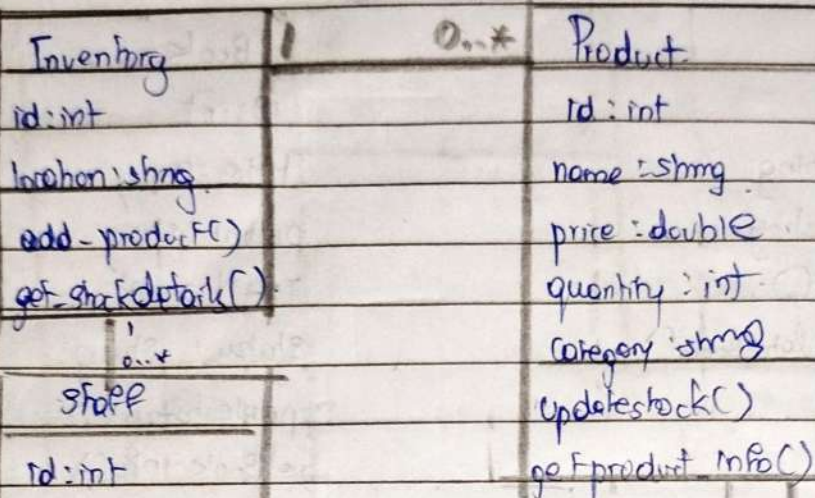
③ Credit Card Processing



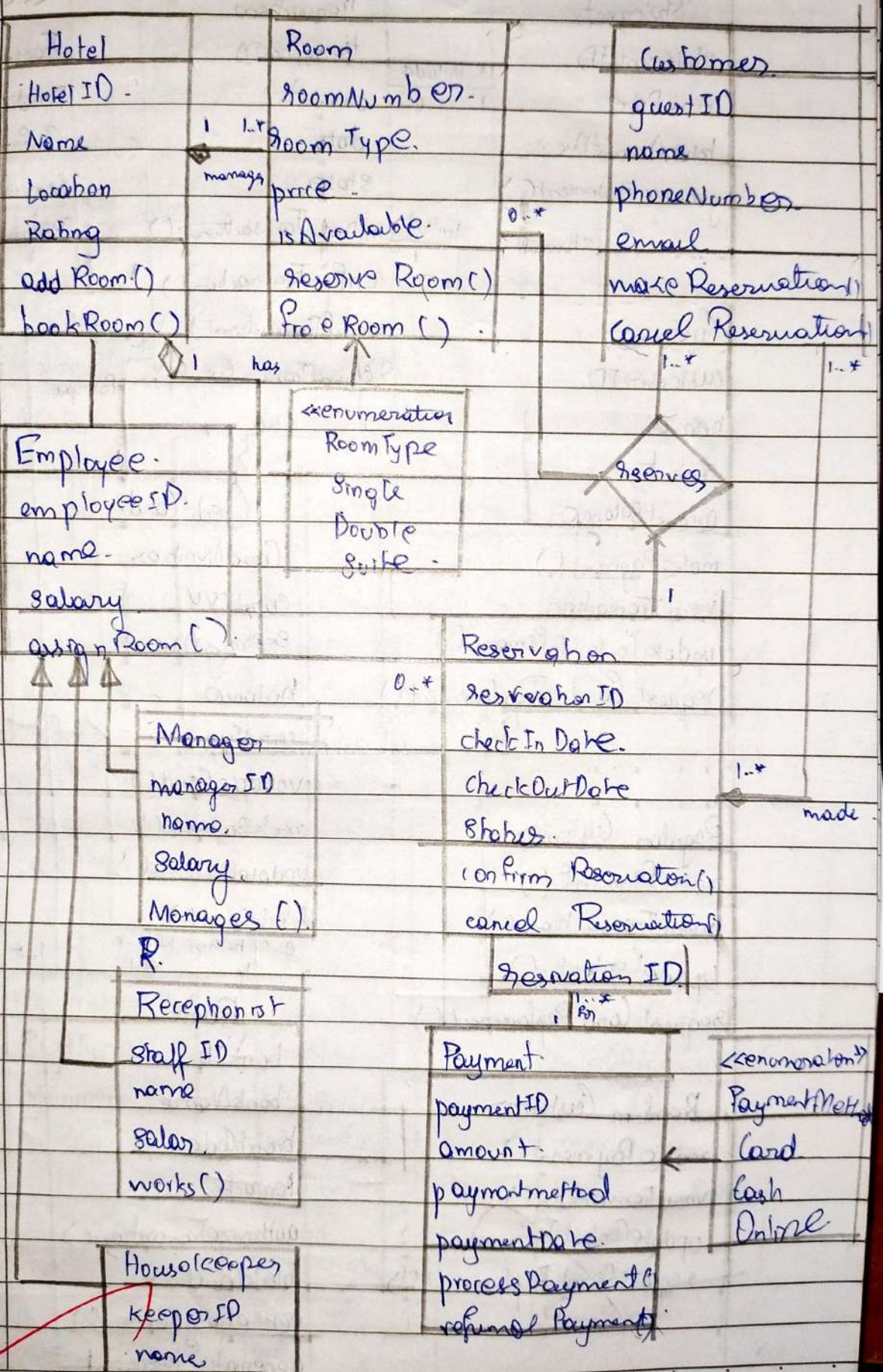
Library management System



⑤ Stock Maintenance System.

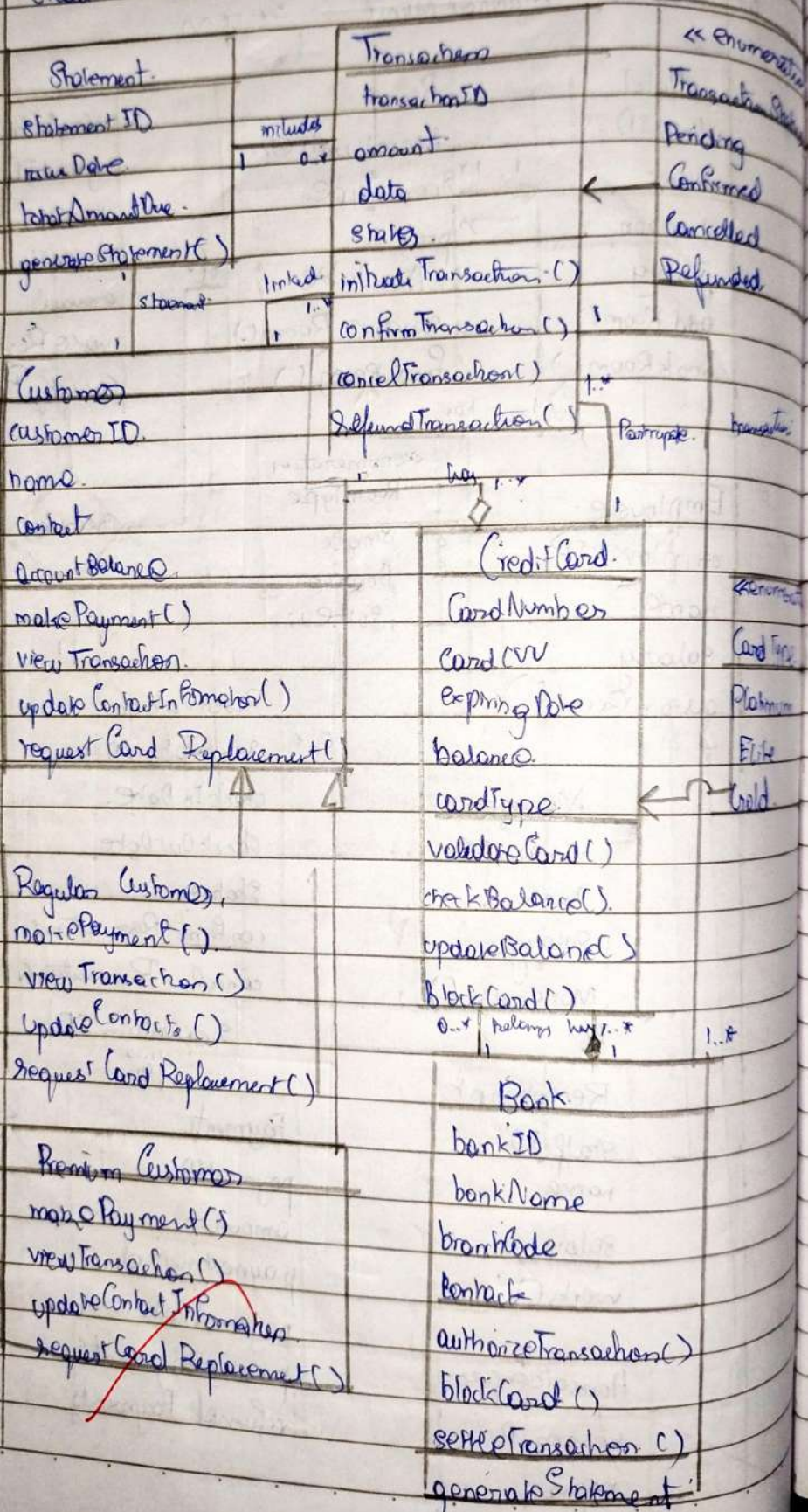


HOTEL MANAGEMENT SYSTEM



Credit Card Processing System

(3)



W/ 12/9

Regular Member

borrowBook()

return^{book}Book()

Premium Member

borrowBook()

returnBook()

Elite member

borrowBook()

returnBook()

Library

Library ID

name

address

totalBooks

addBook()

removeBook()

manages

Books

ISBN

1st title

contains

author

publicationYear

borrowBook()

returnBook()

ISBN

Member

member ID

name

Contact

membership

borrowBook()

returnBook()

Librarian

librarian ID

name

Contact

workshift

manageInventory()

assistMember()

BorrowTransaction

transaction ID

borrowDate

returnDate

fineAmount

calculateFine()

Fine

fine ID

fineAmount

status

calculateFine()

payFine()

<<enumeration>>

MembershipType

Regular

Premium

Elite

Reservation

1st reservation ID

reservedDate

status

reserveBook()

cancelReservation()

<<enumeration>>

ReservationStatus

Active

Cancelled

Completed

<<enumeration>>

ShiftType

Morning

Evening

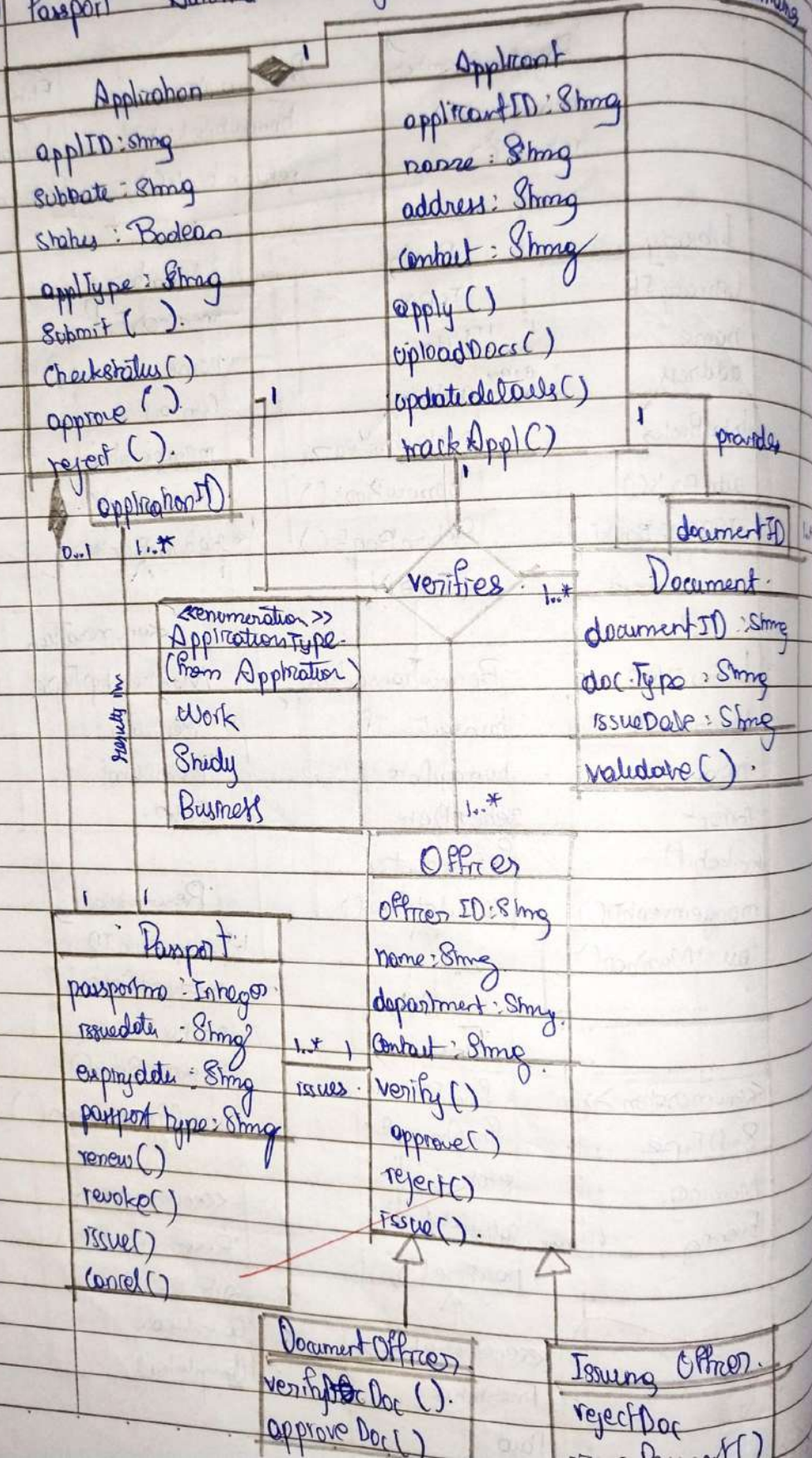
<<enumeration>>

FineStatus

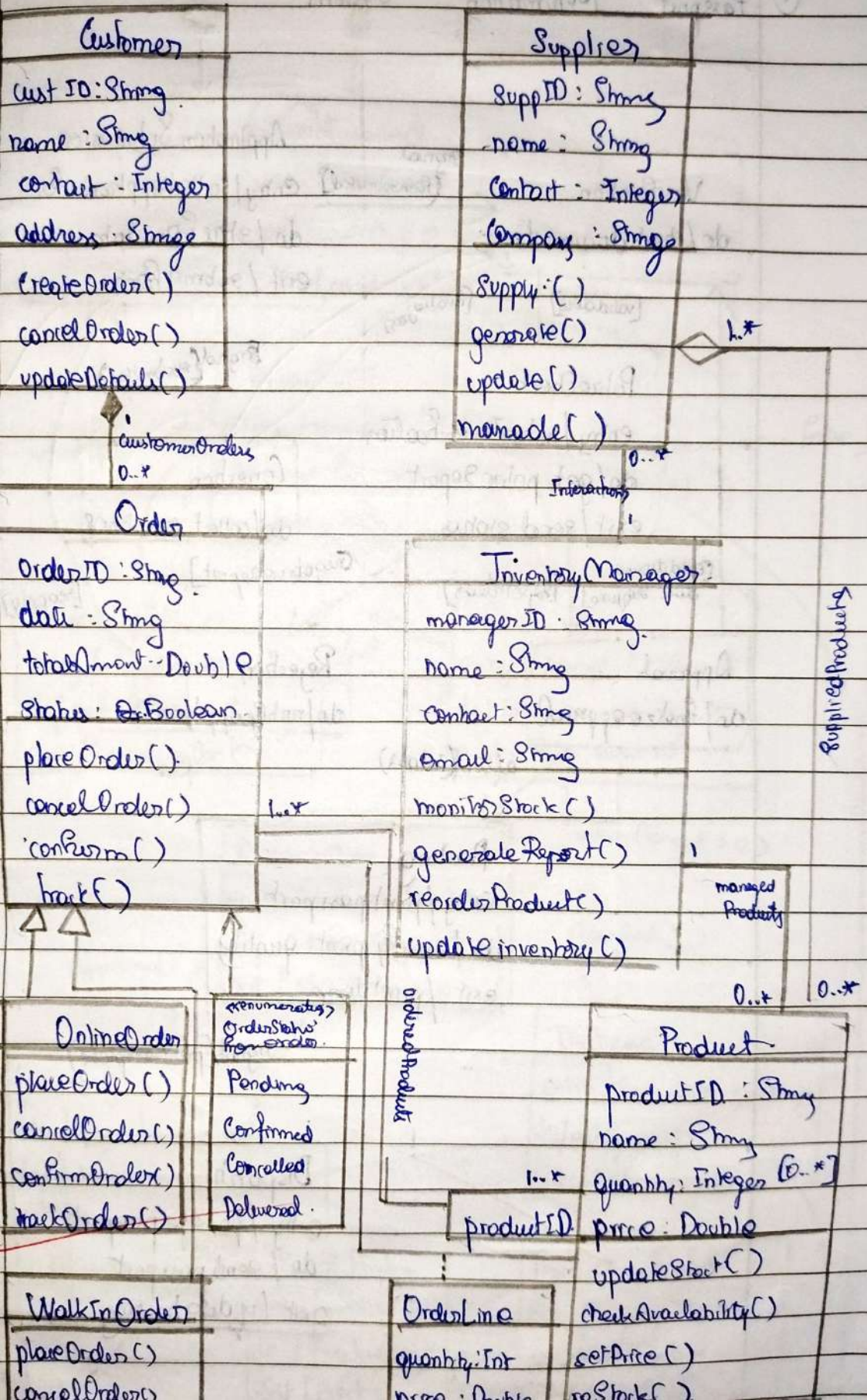
Paid

imposed

④ Passport Automation System



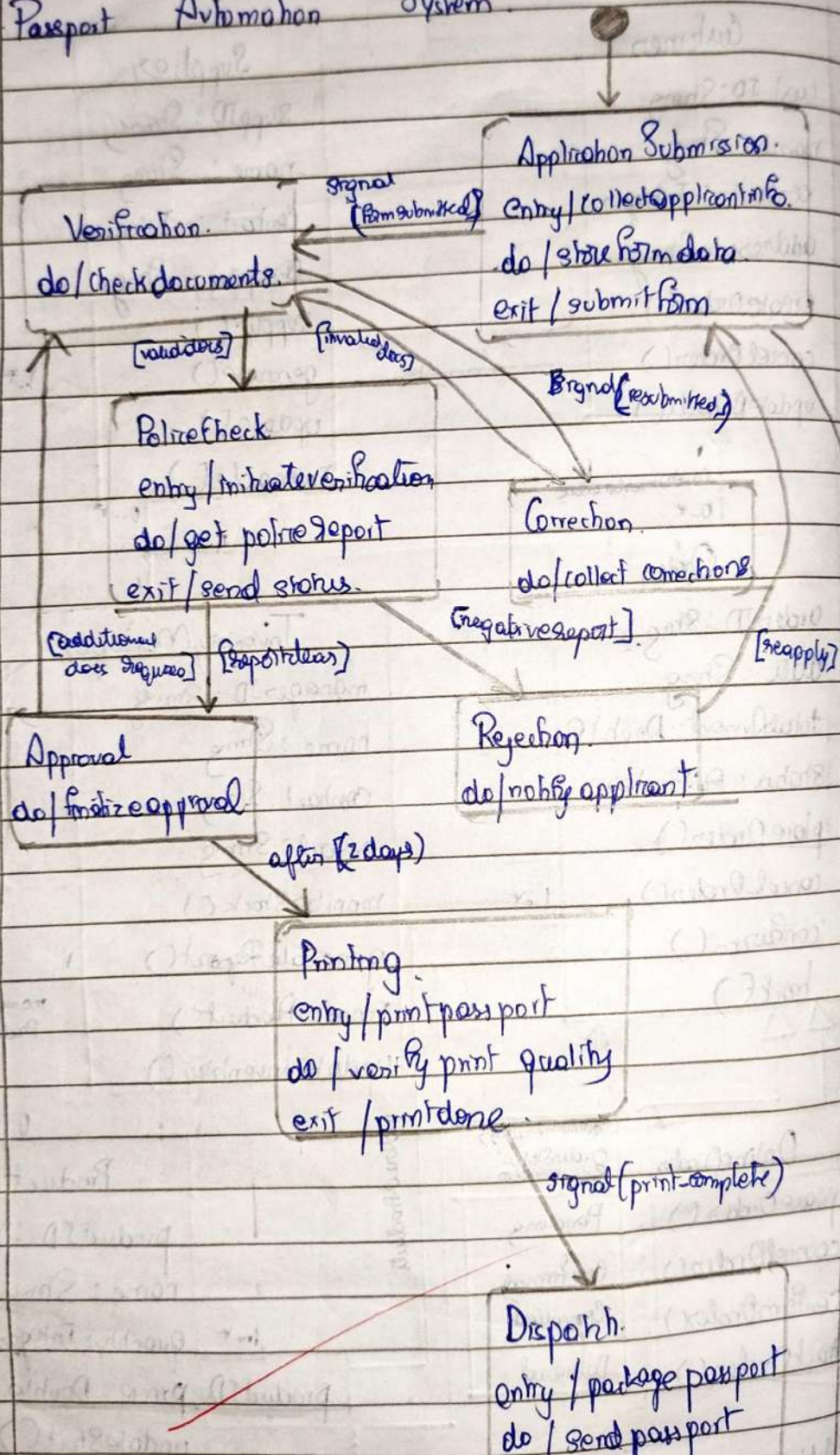
Stock Maintenance System



STATE CHART DIAGRAM

Date 10/10/25
Page

① Passport Automation System



CardInserted

entry / detect card

do / read chip or swipedata

exit / card validated

signal (card-detected)

PinVerification

when (pin-correct == true)
entry / prompt for PIN
do / wait for input
exit / store PIN

when (pin-correct == false)

TransactionSelection

entry / show options

do / select transaction type

exit / record selection

signal (option selected)

AmountEntry

entry / prompt for amount

do / take amount input

exit / save amount

when (amount > 0)

Authorization

entry / send details to bank

do / wait for response

exit / store response

[rejected]

approved

Processing

entry / begin transaction

do / debit amount

exit / record success

offer (3s)

Receipt Printing

entry / print slip

do / finalize logs

exit / eject card

signal (print done)

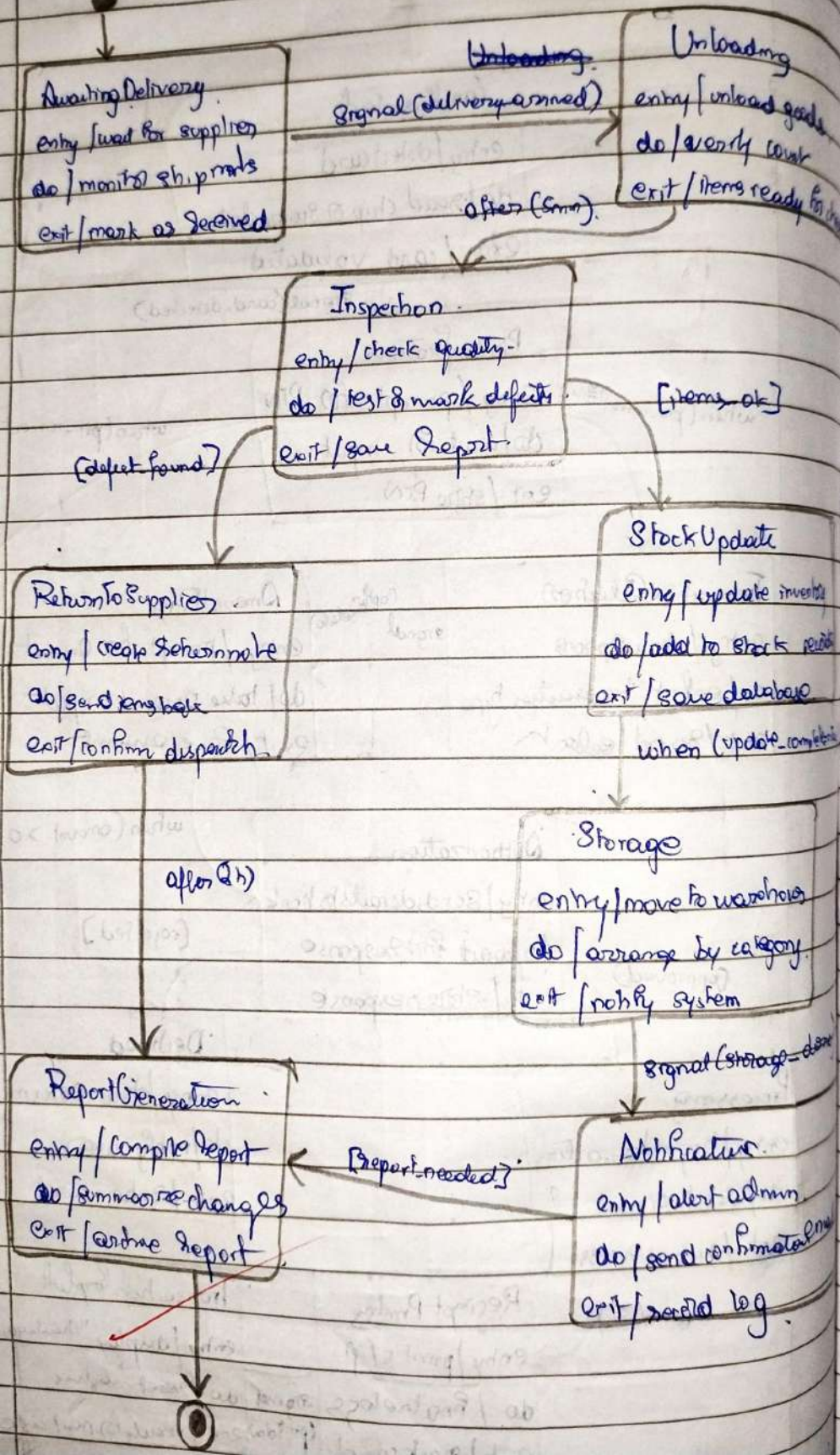
Transaction Complete

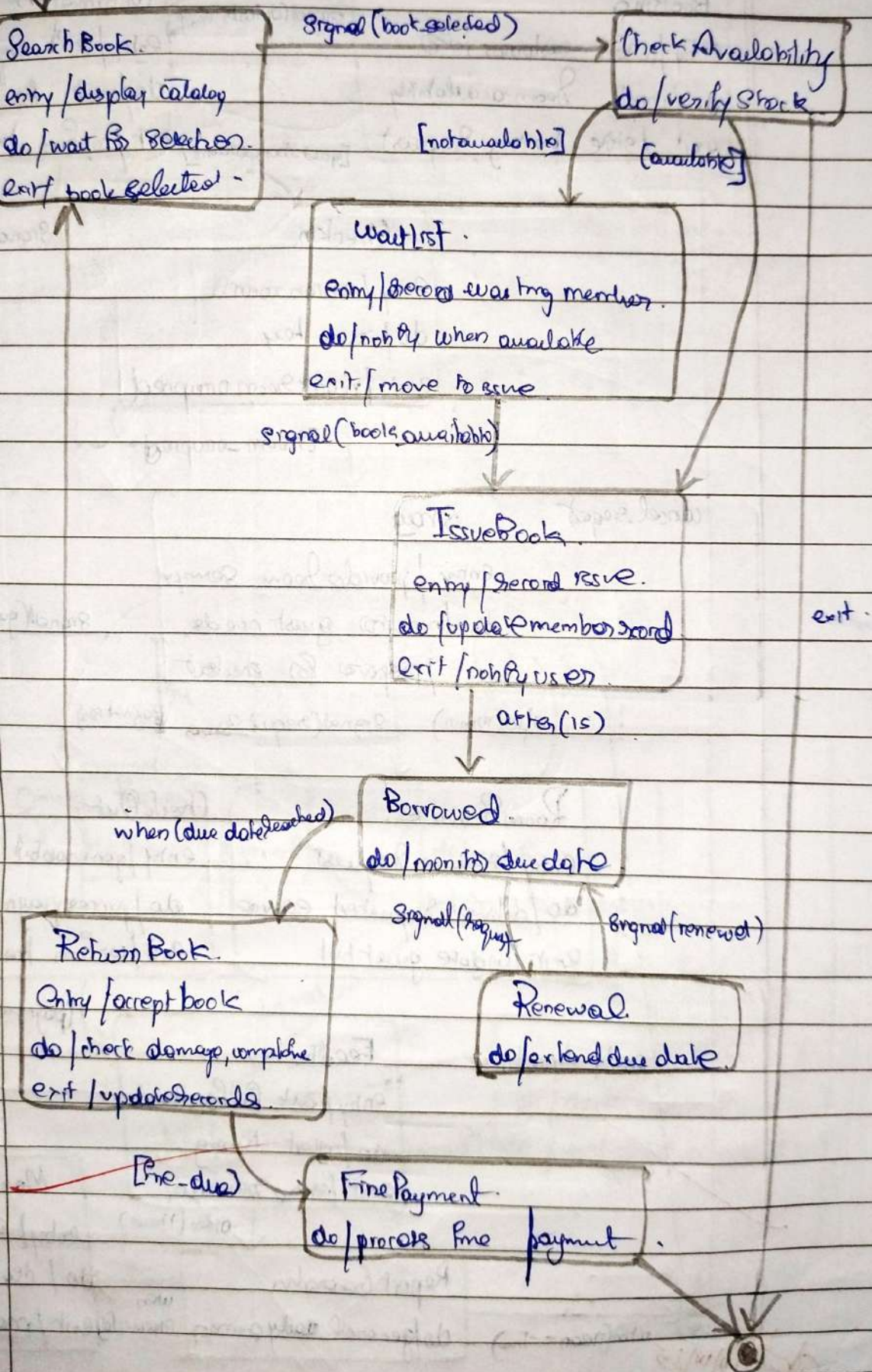
entry / display "Thank you"

do / reset machine

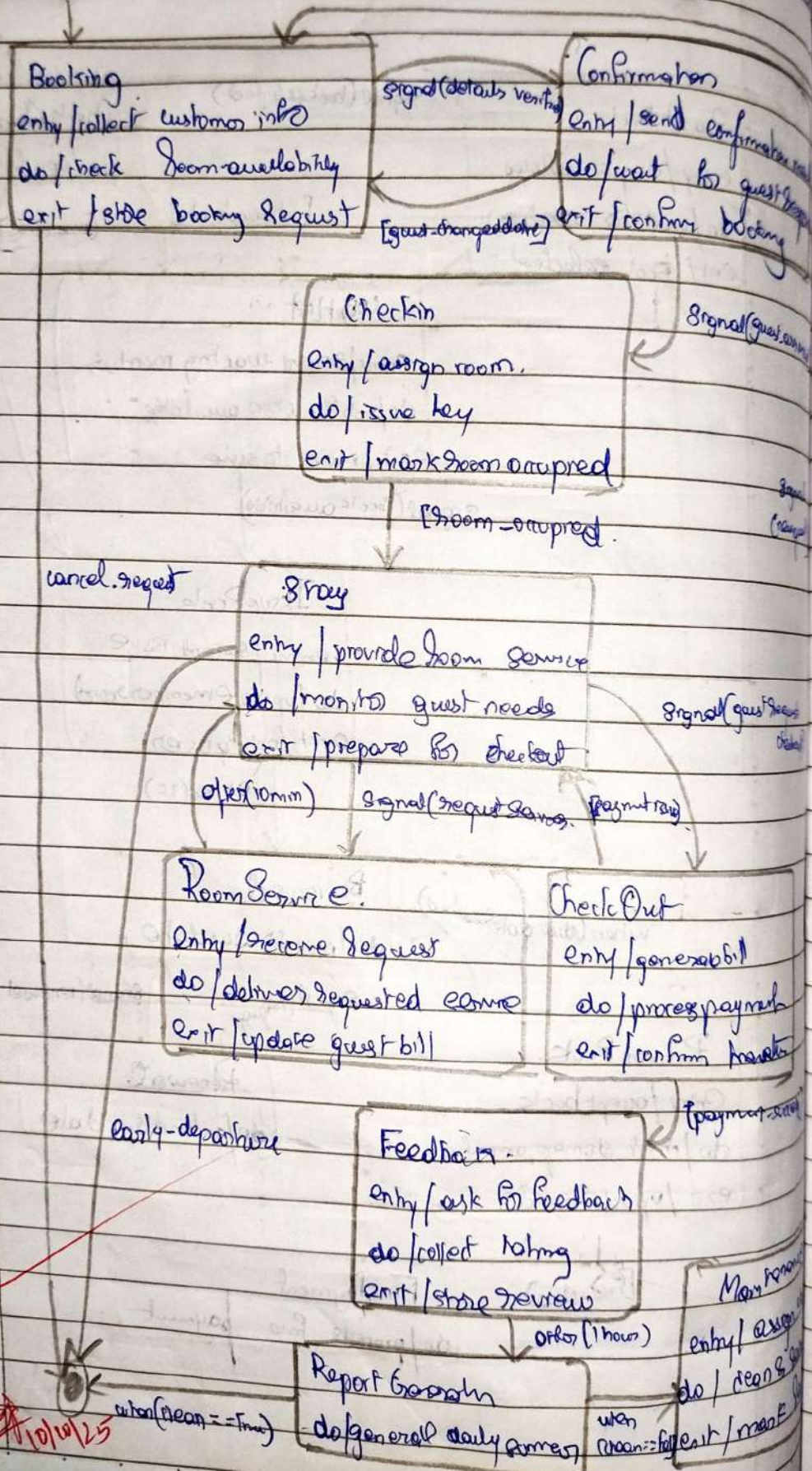
exit / ready for next user

Stock Maintenance System (Restocking ^{control} Flow)





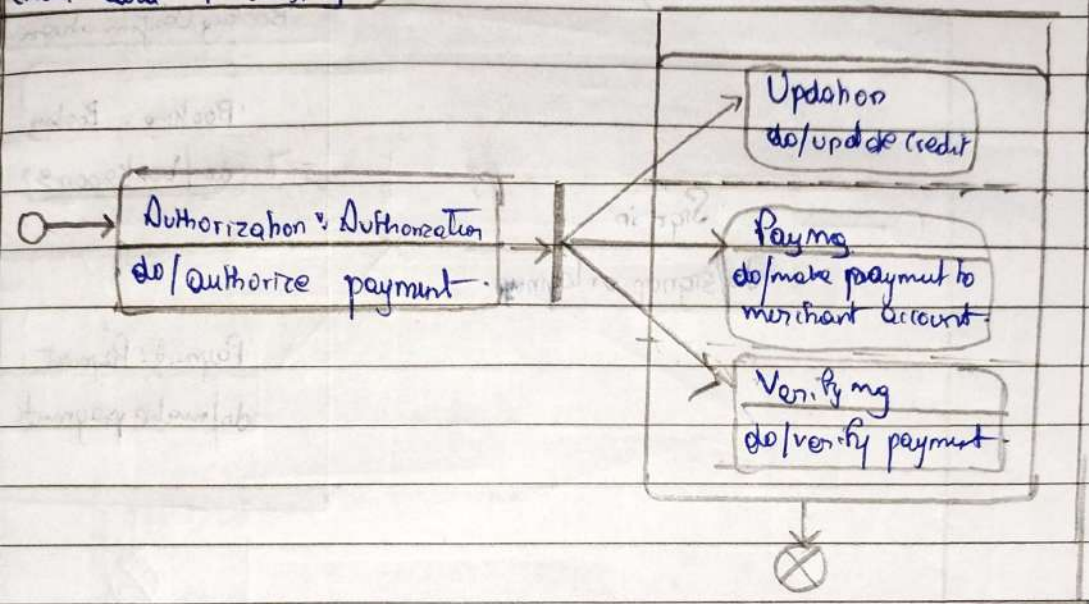
⑤ Hotel Management System (Room booking, checkin, checkout)



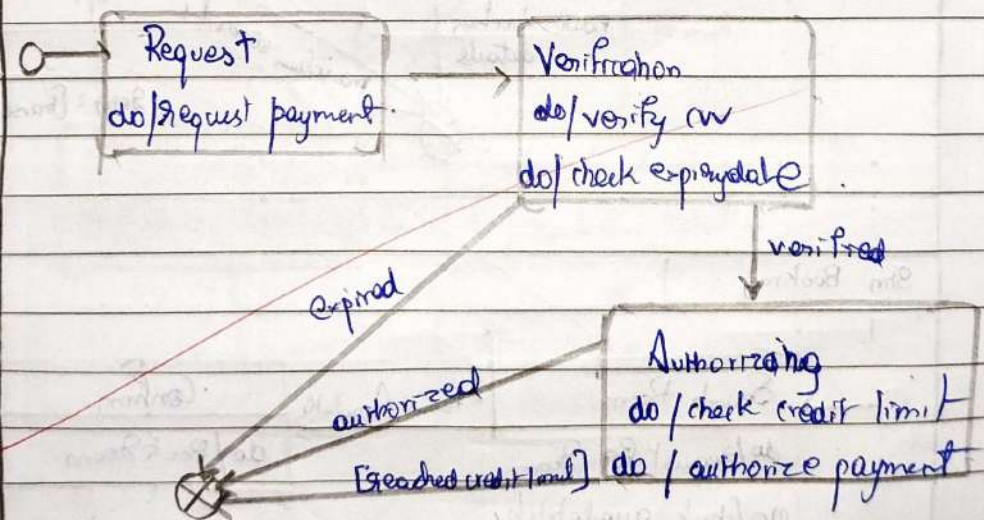
10/10/25

Credit Card Processing

Credit Card Processing

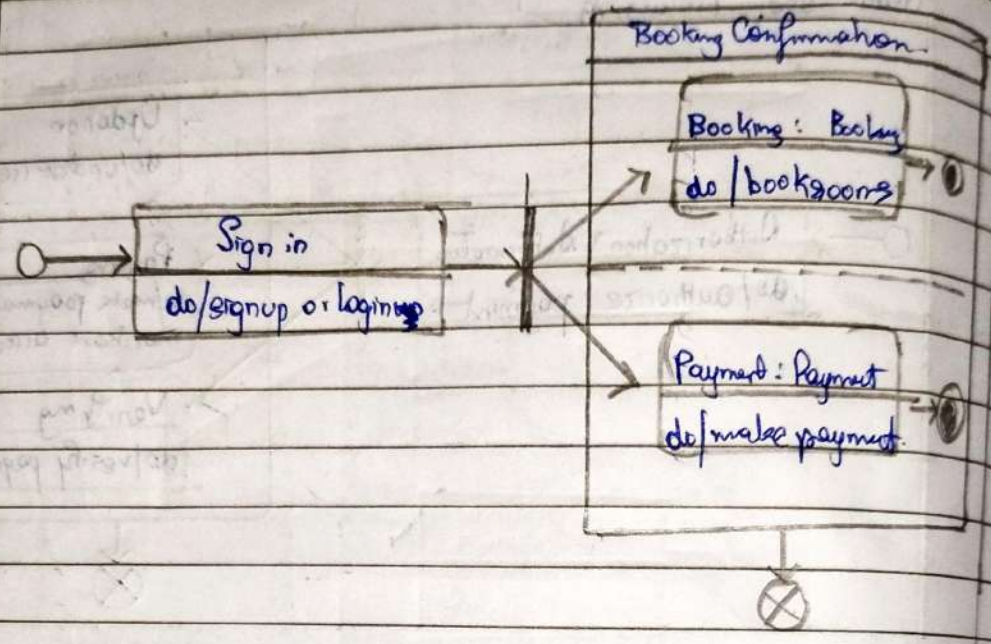


sfm Authorization

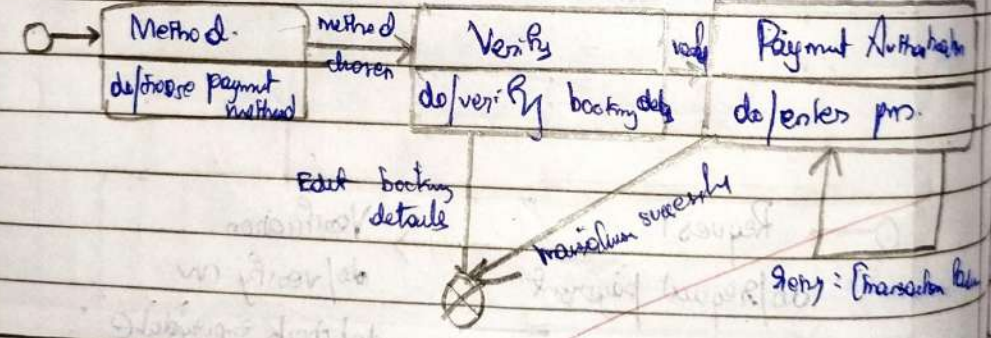


Hotel Management System

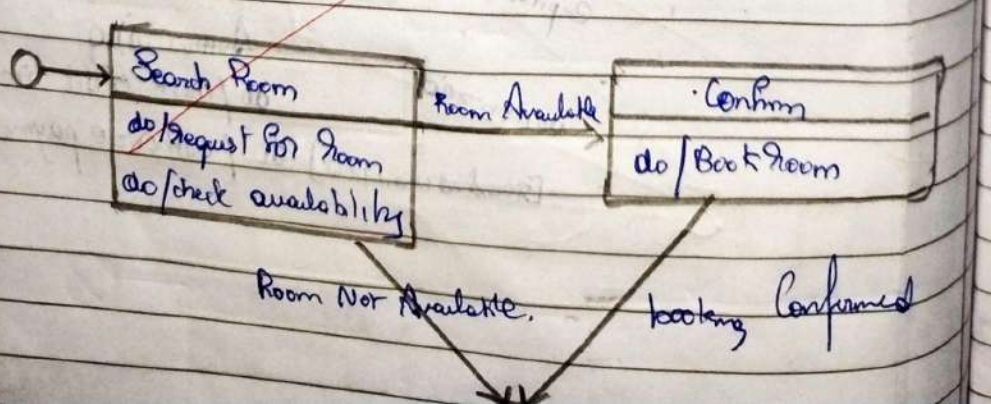
sm Hotel Management



sm Paymt

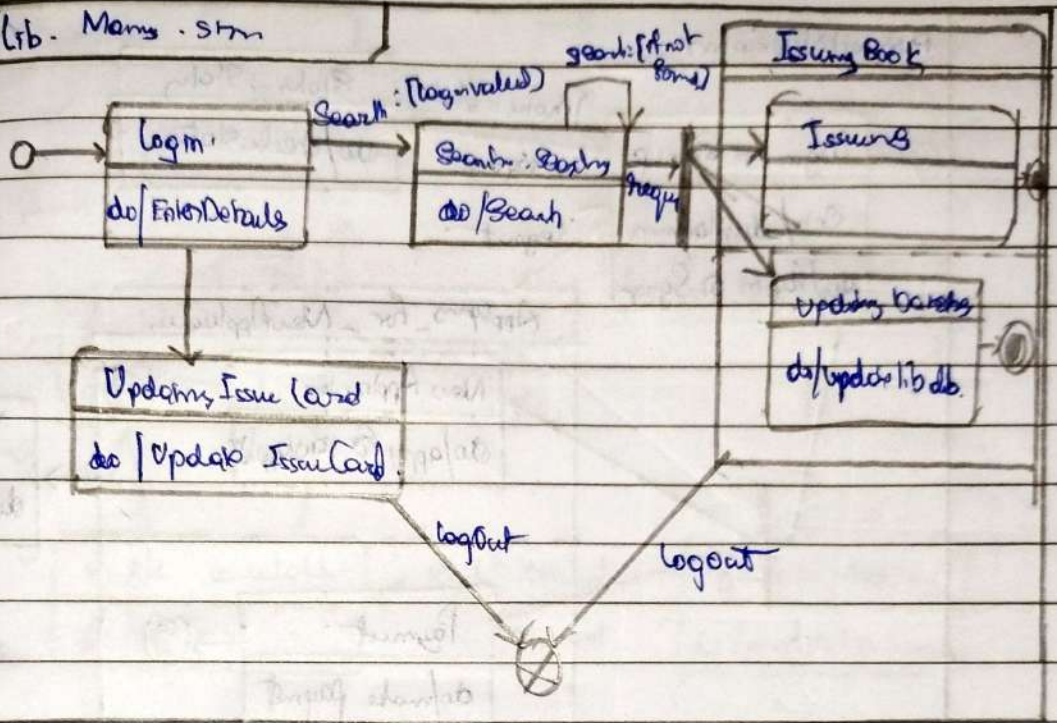


sm Booking

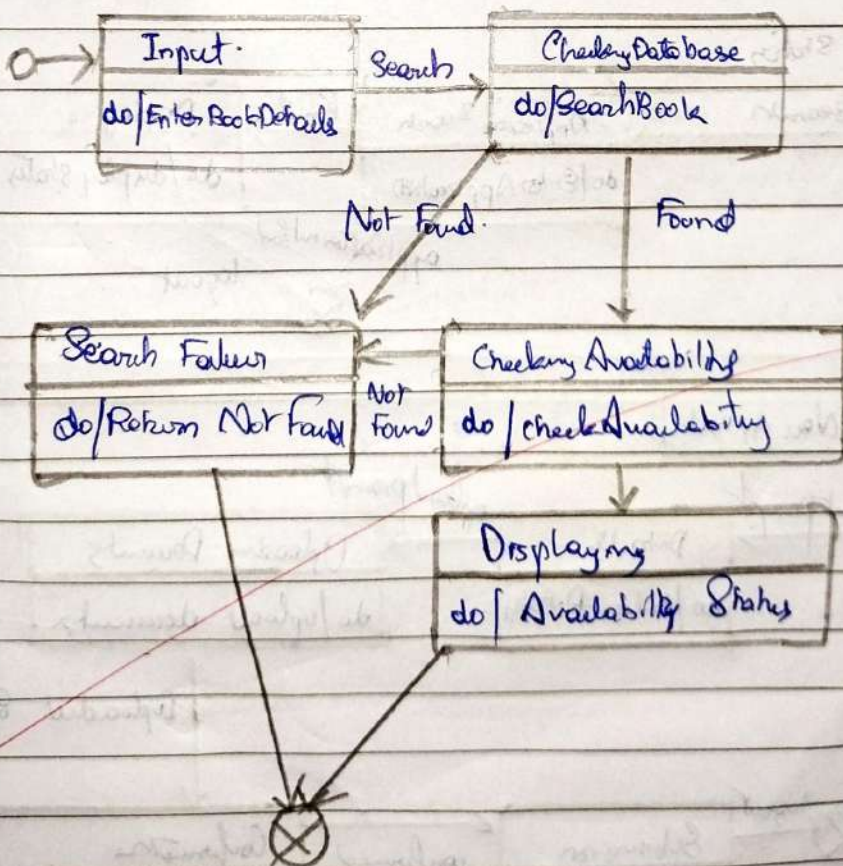


Library Management System

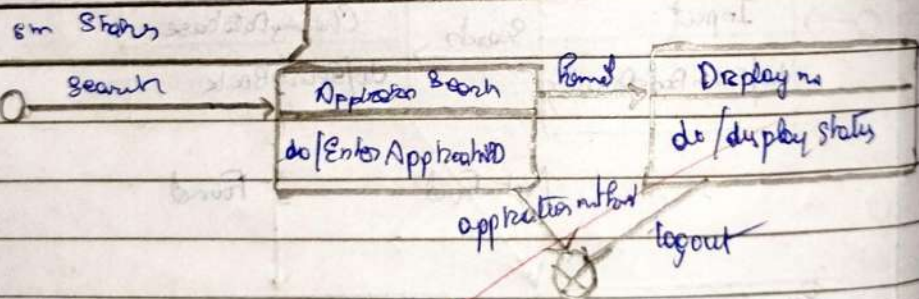
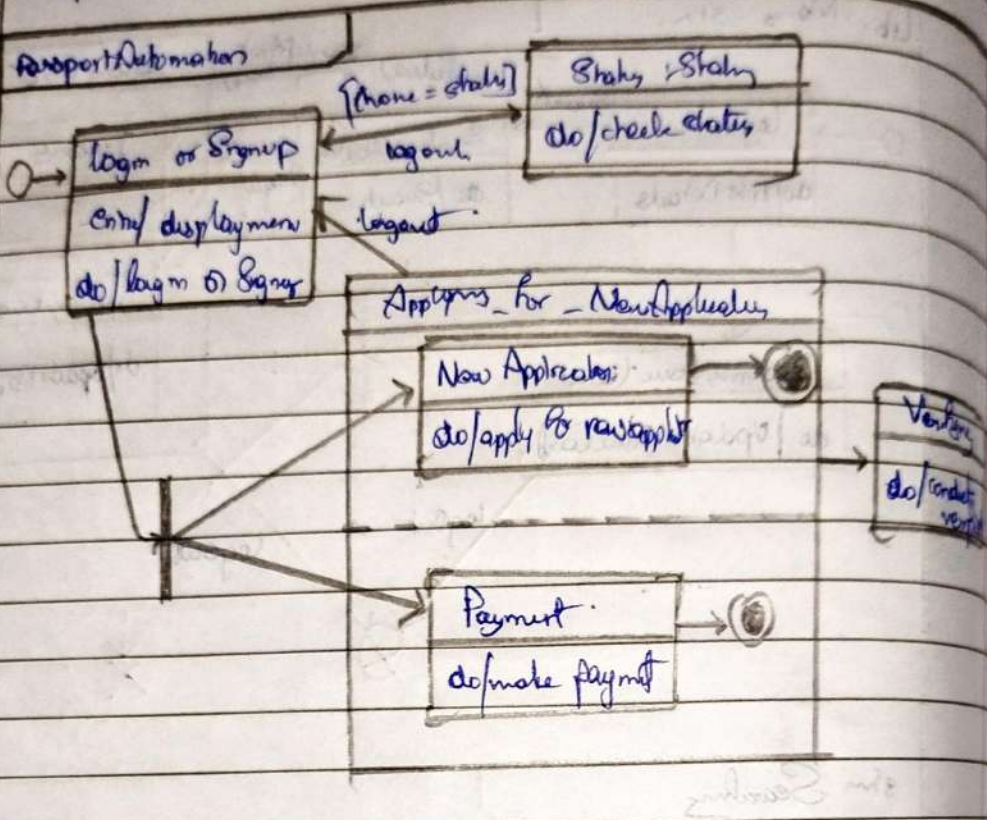
Lib. Mgmt. ssm



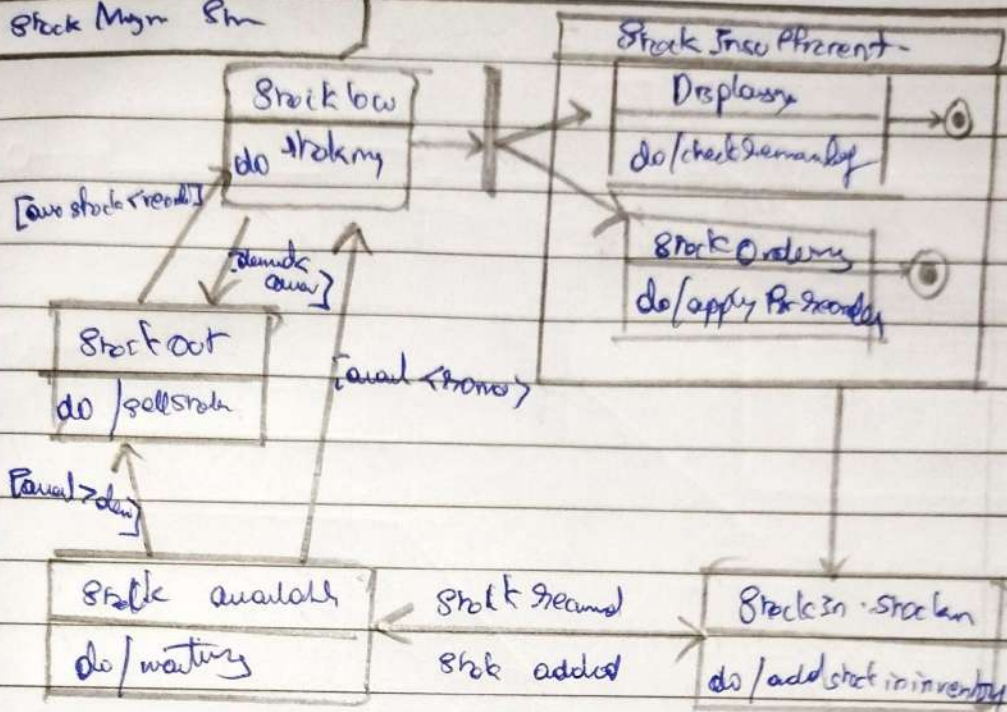
ssm Searching



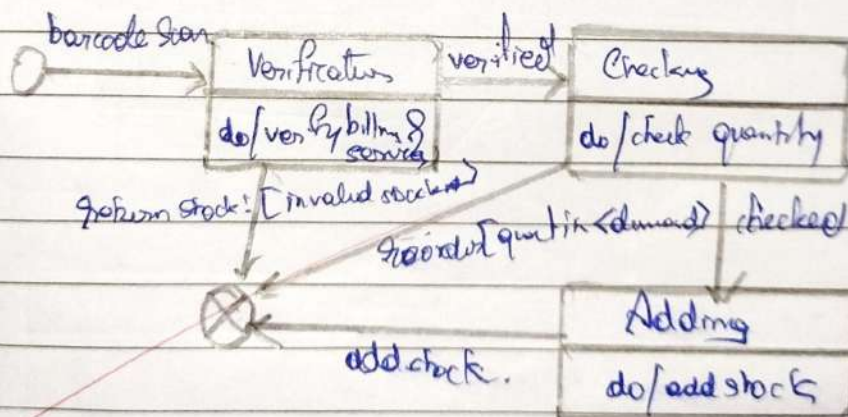
④ Passport Automation System



Stock Mgmt Sm



arm Stocking



Credit Card Processing System



Payment Gateway



Bank Staff



Administrator

Process Bill Payments

Authorize Transaction

Handle Refunds and Cancellations

Validate Card Details

View Transaction History Reports

Manage Credit Limits and Balances

Maintain Customer Profile

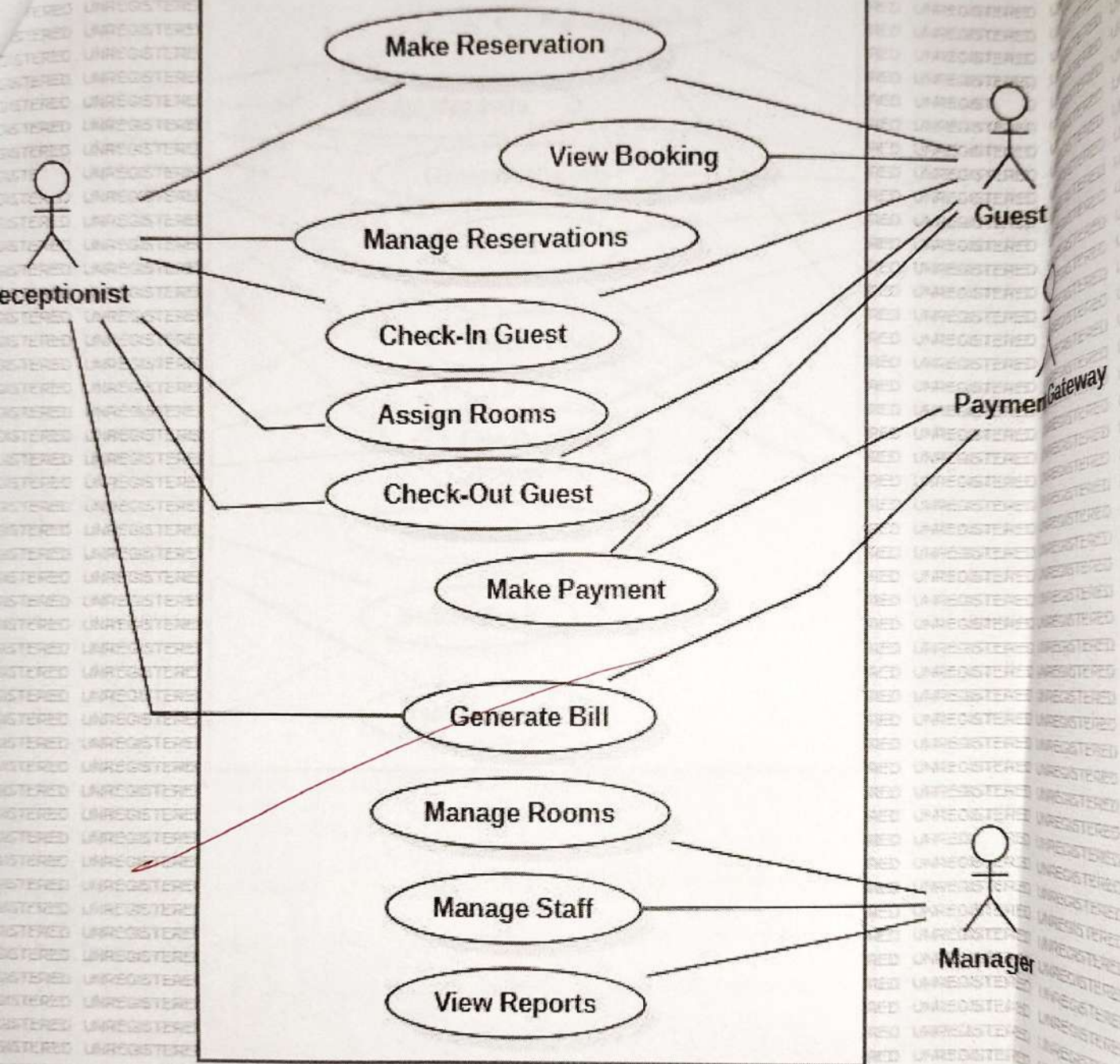
Generate Monthly Bills

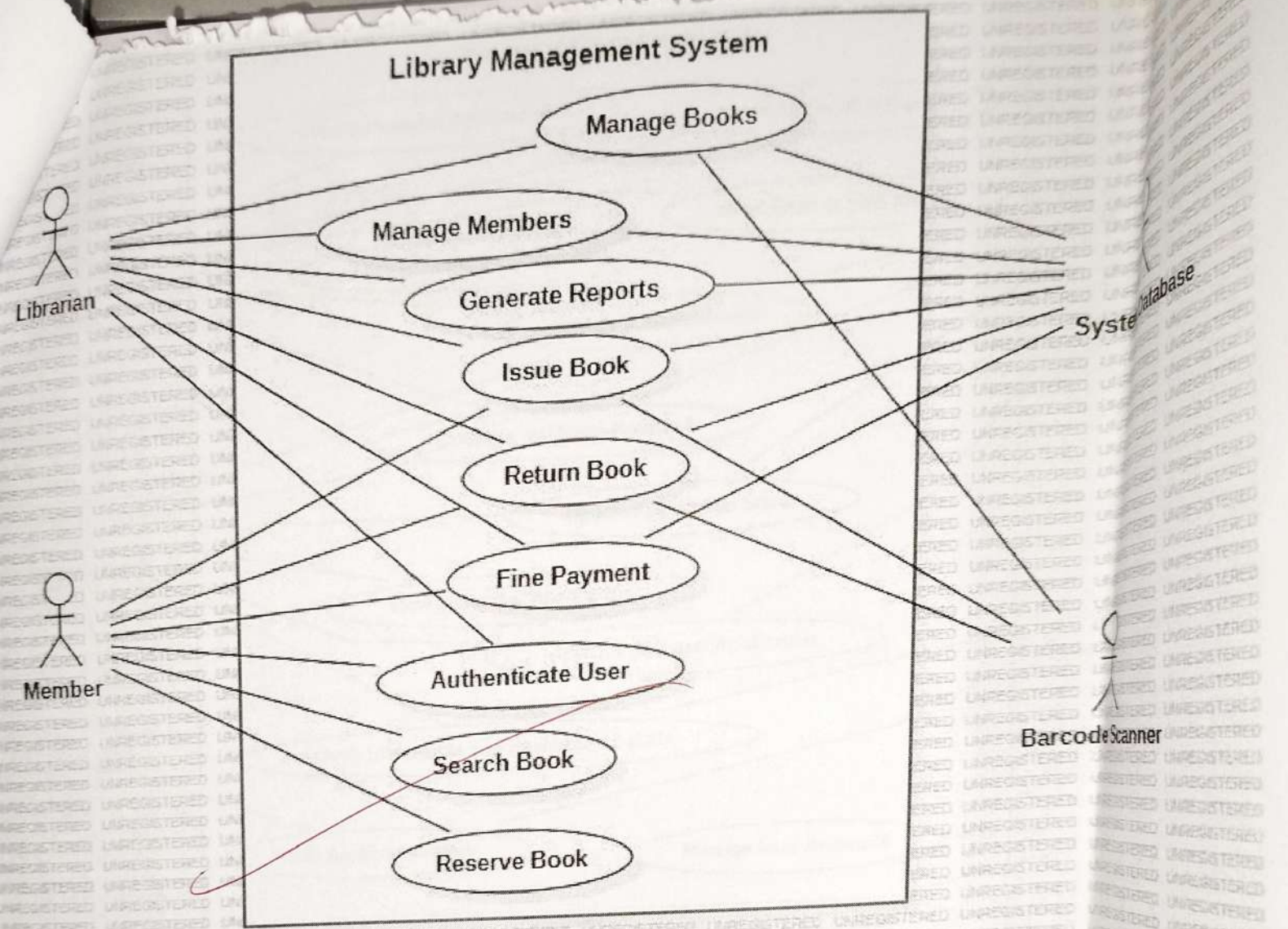
Alert Administrator

Detect Fraudulent Transactions

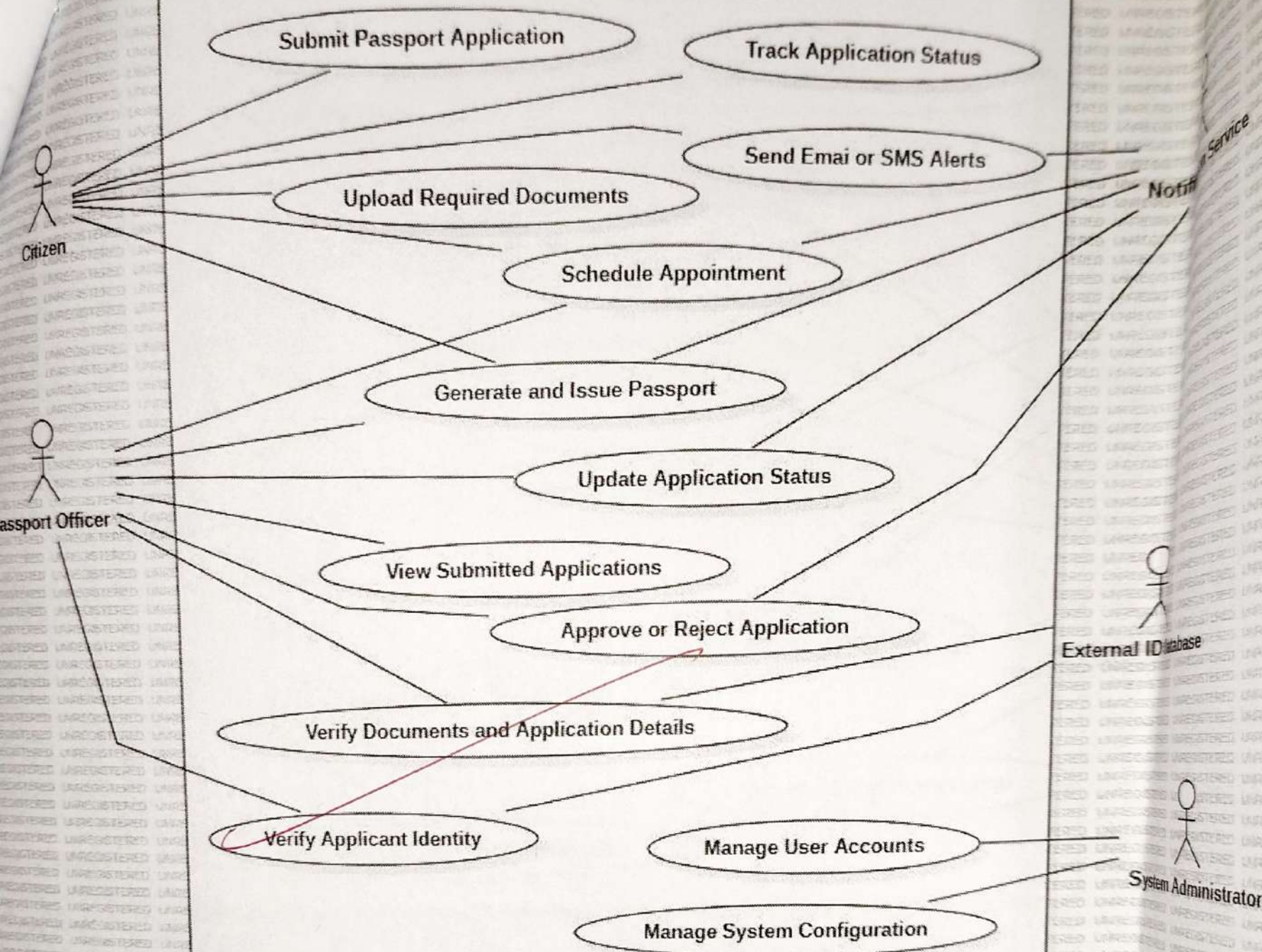
[View Reports](#)

Hotel Management System

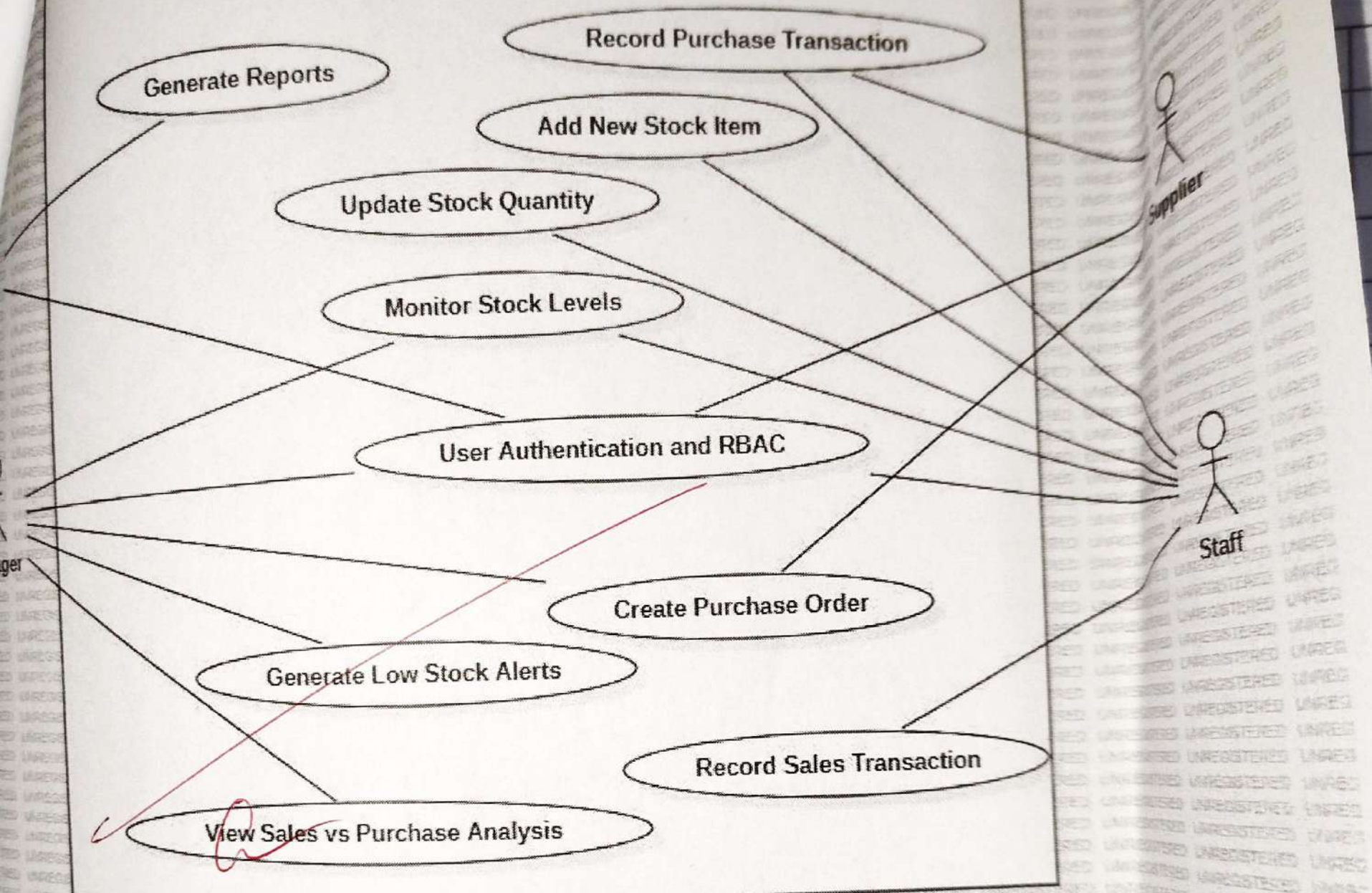




Passport Automation System



Stock Maintenance System



Authorize Transaction (Successful)

Customer

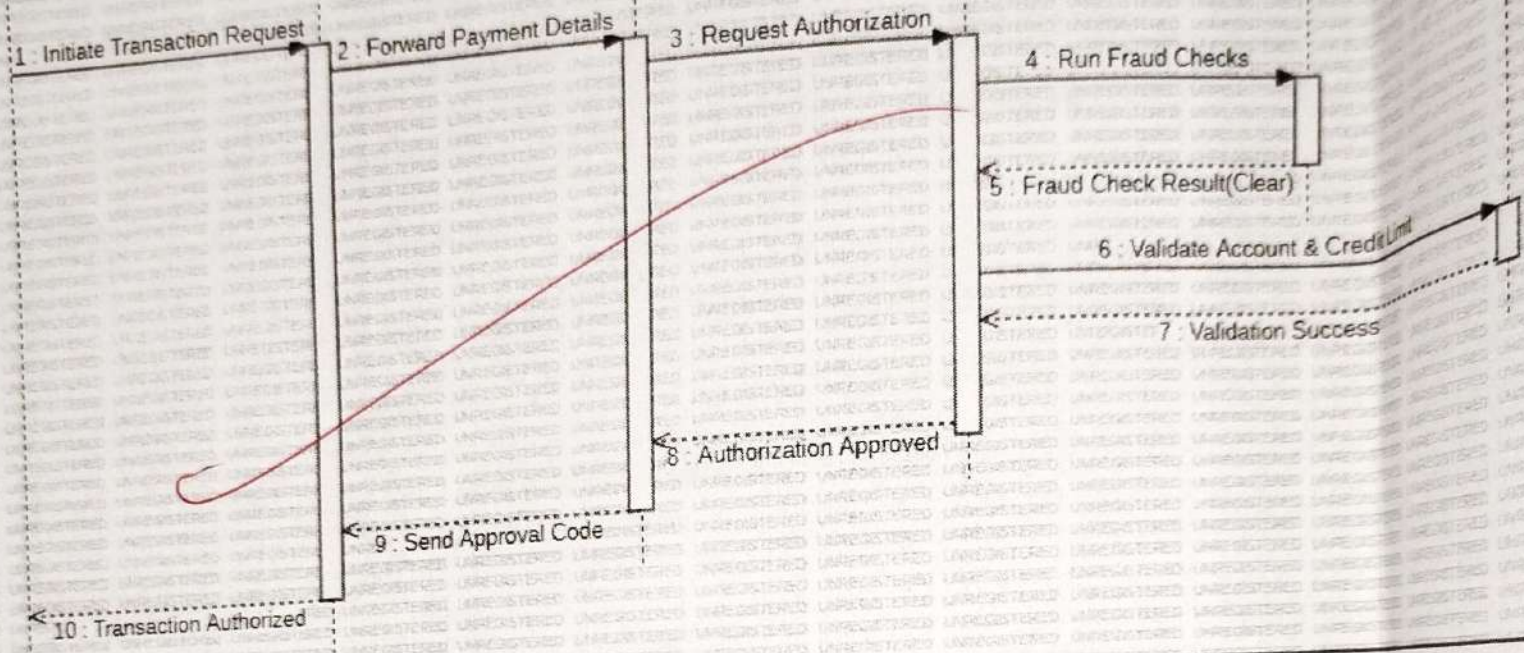
Merchant

Payment Gateway

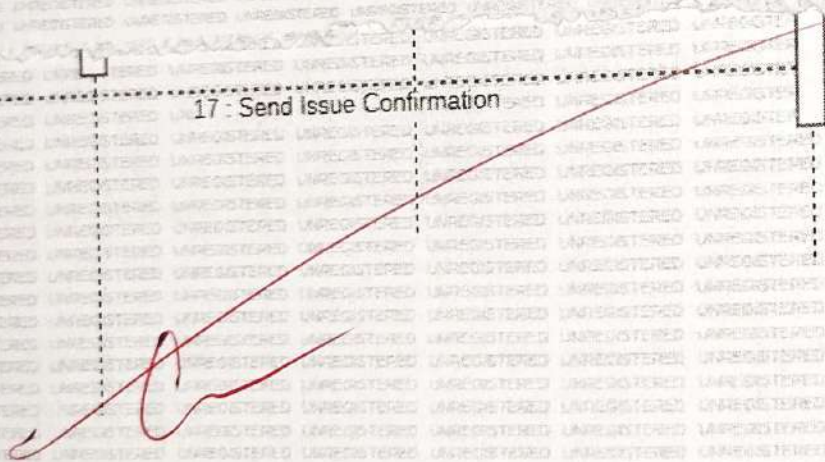
Authorisation Server

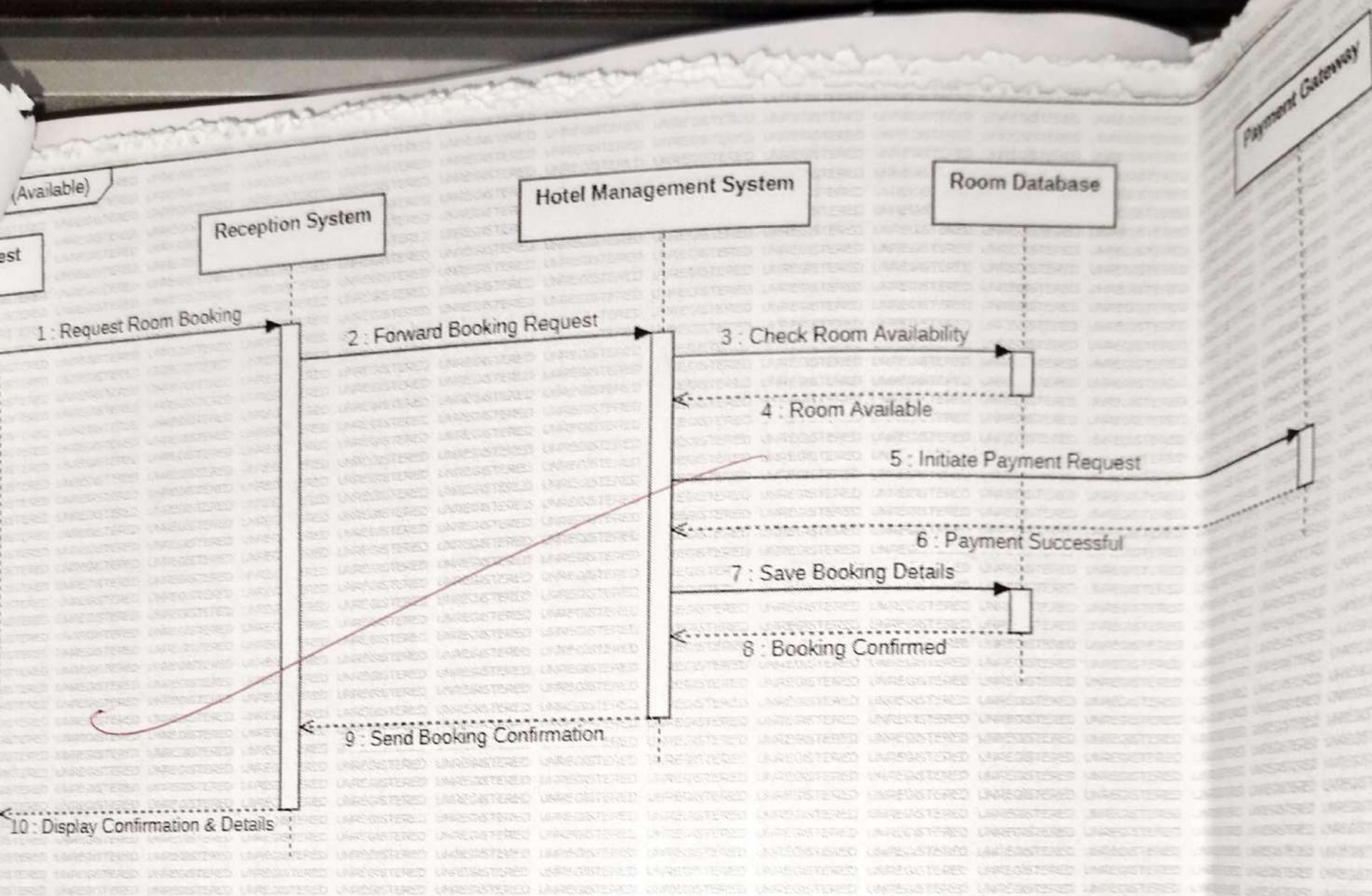
Fraud Detection Engine

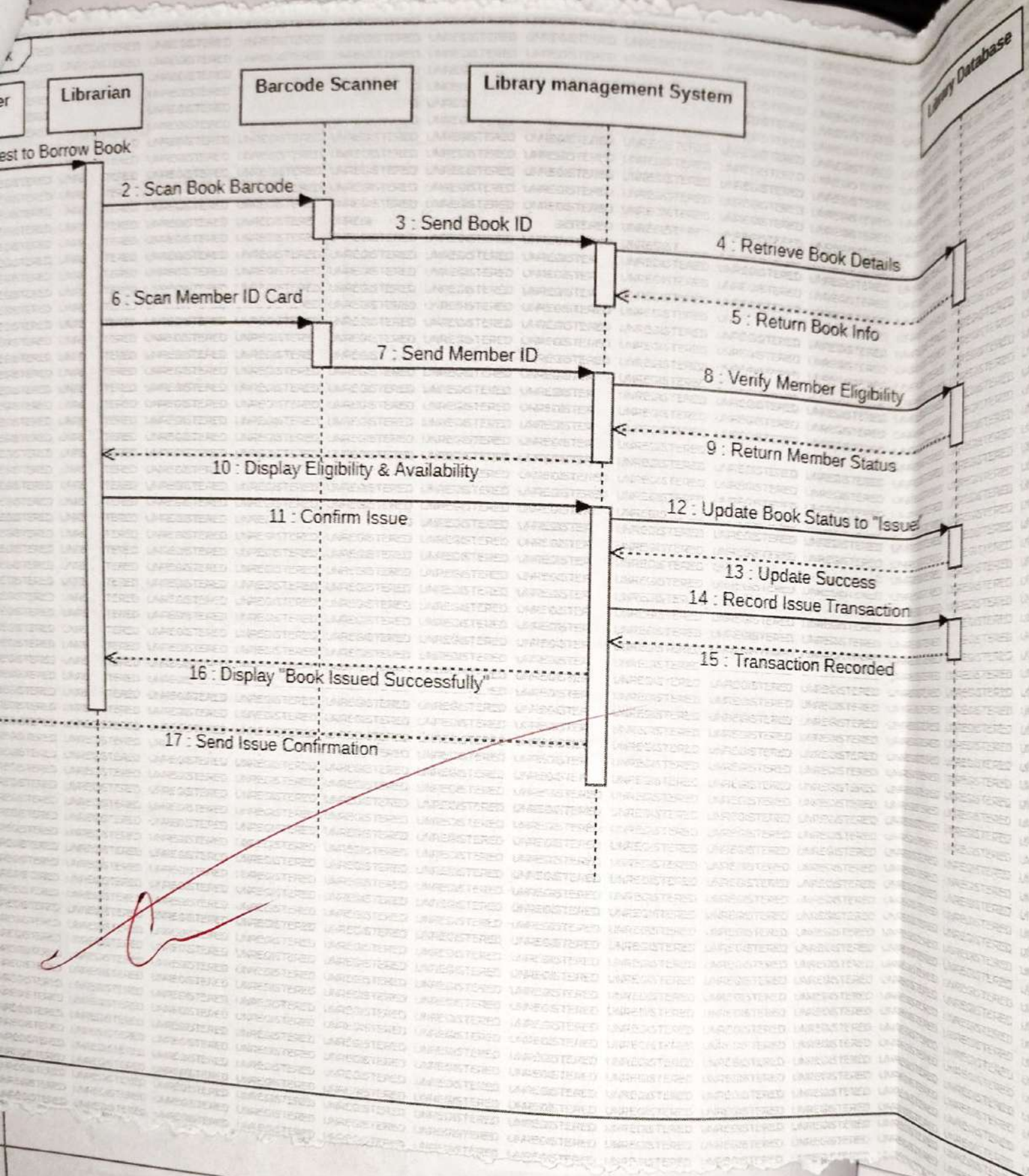
Bank Database



17: Send Issue Confirmation





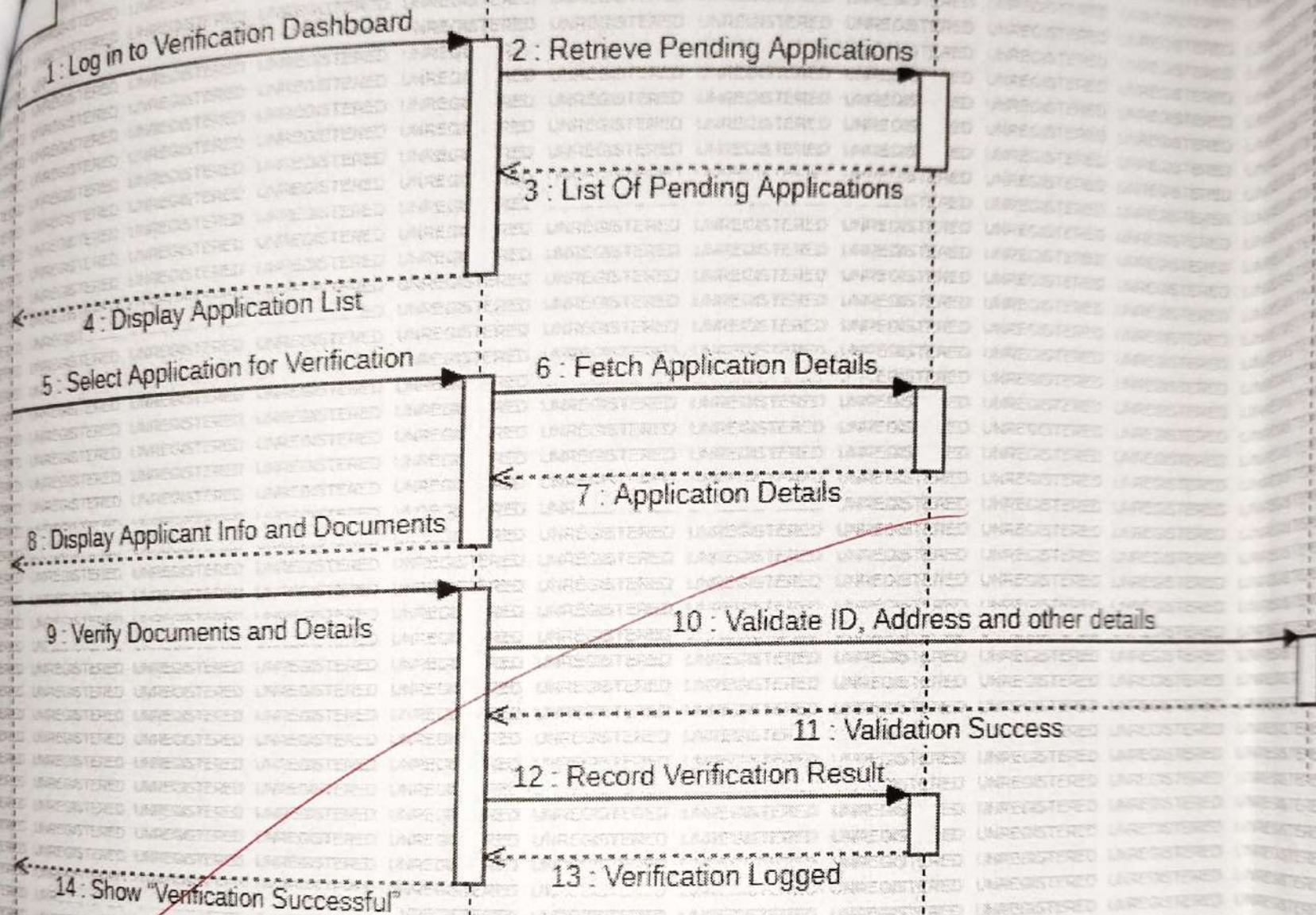


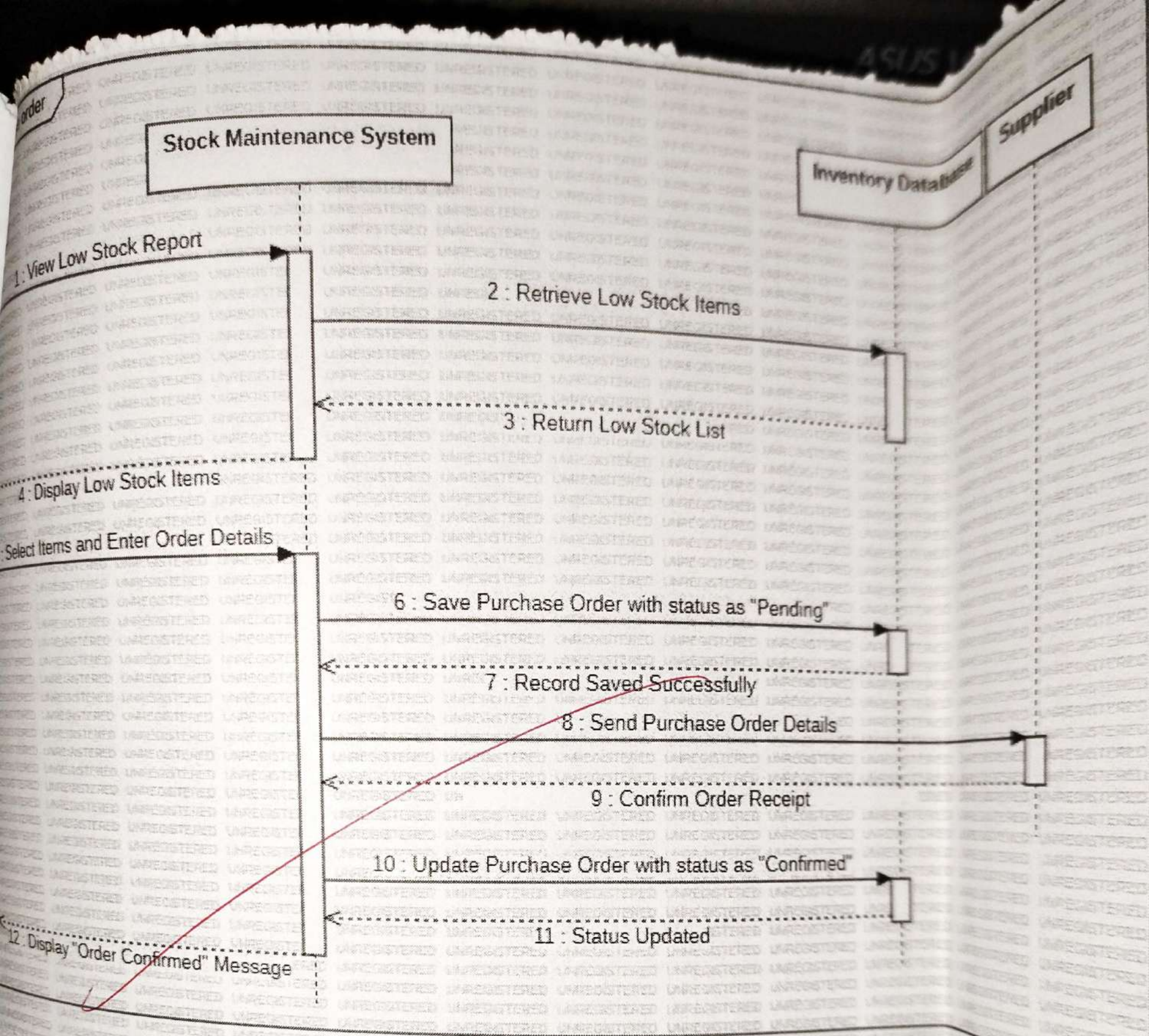
Application (Successful Verification)

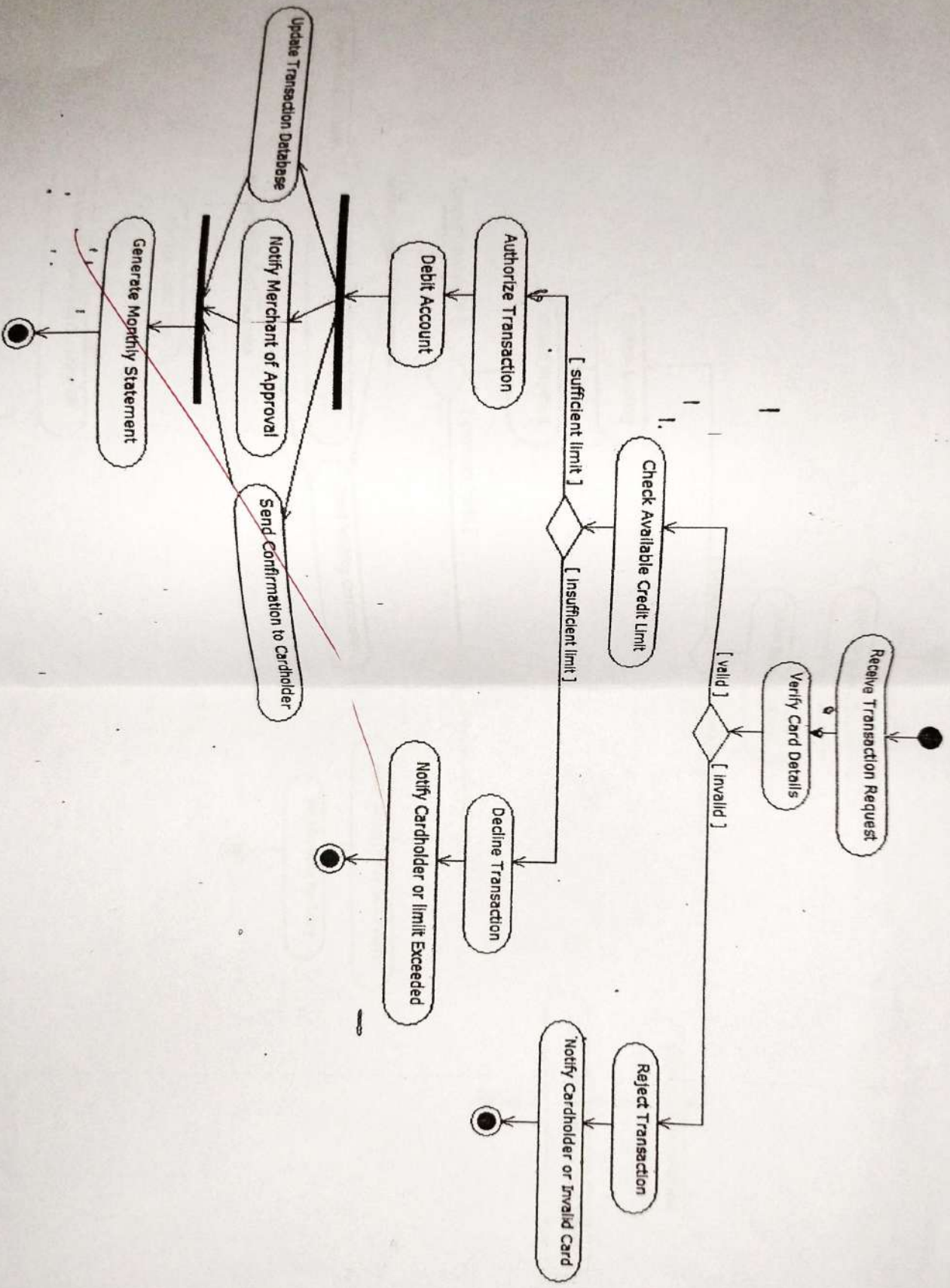
Application System

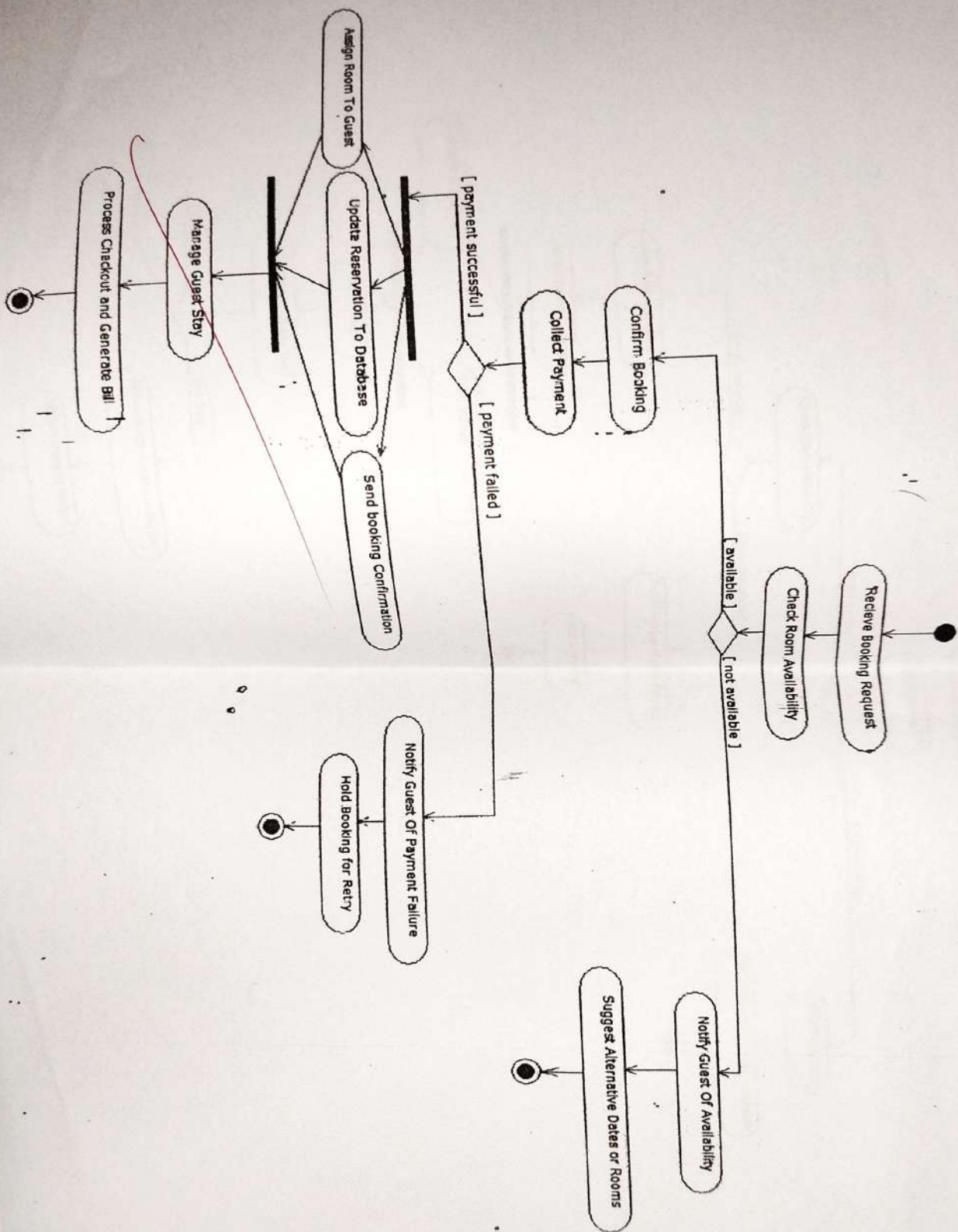
Application Database

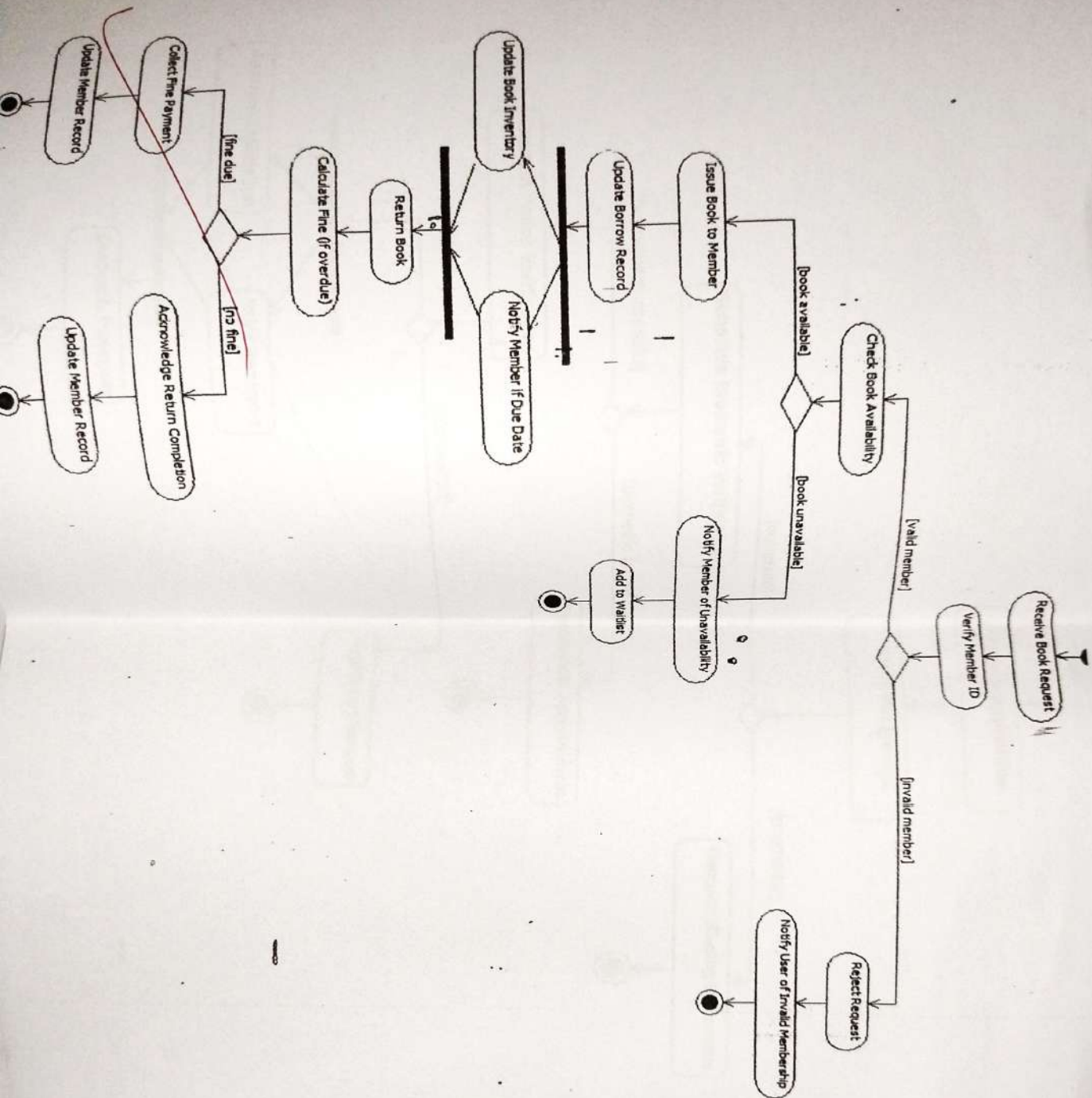
External ID Database

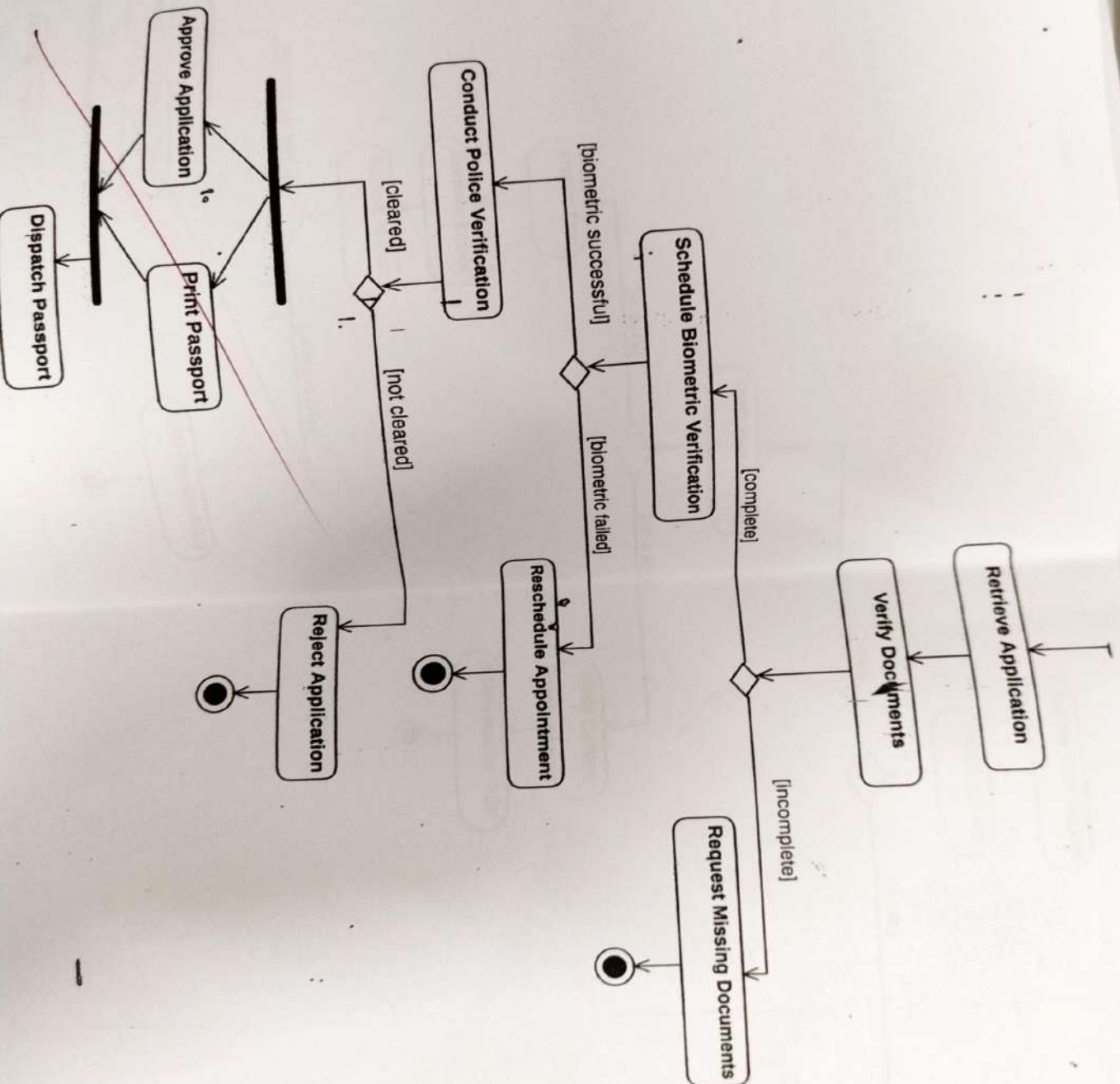


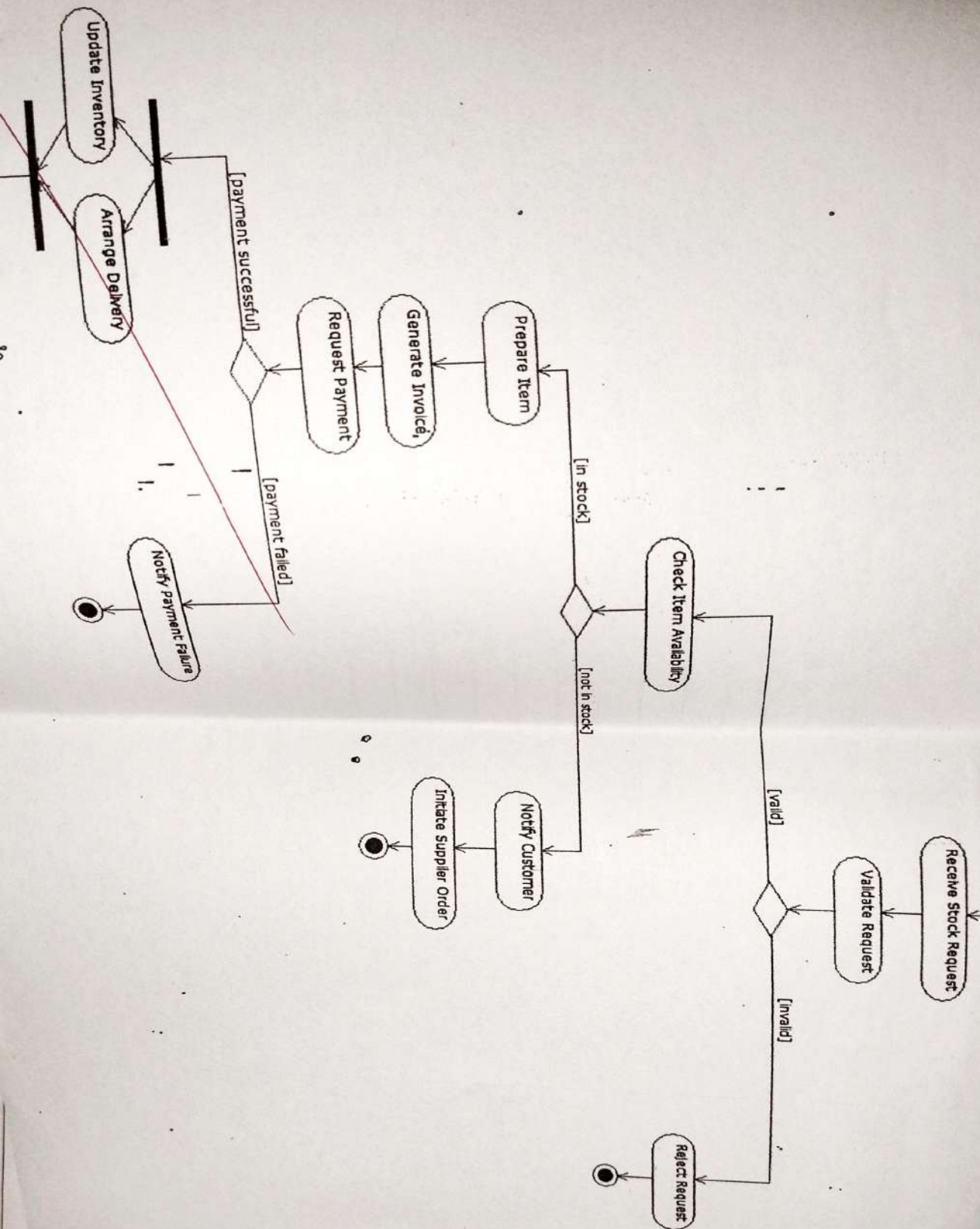


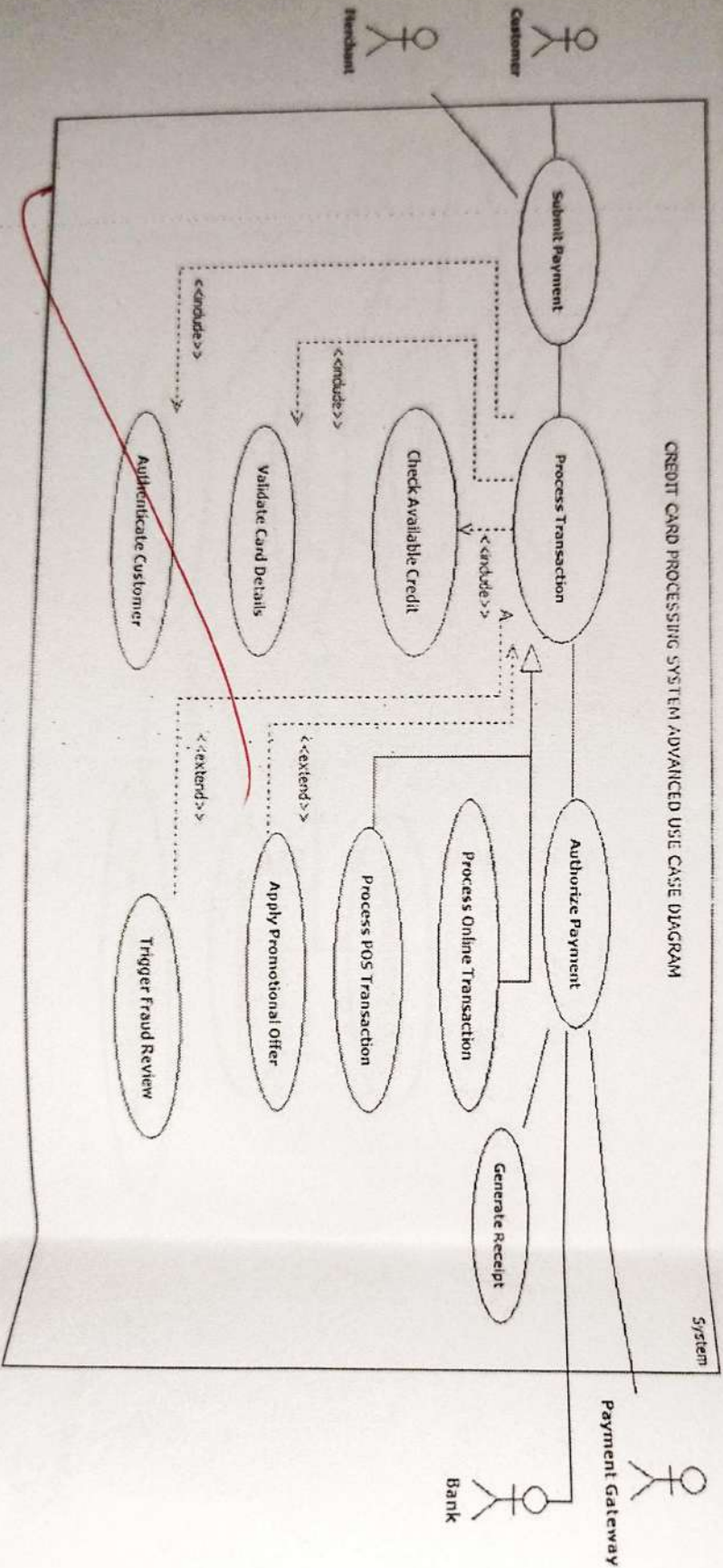






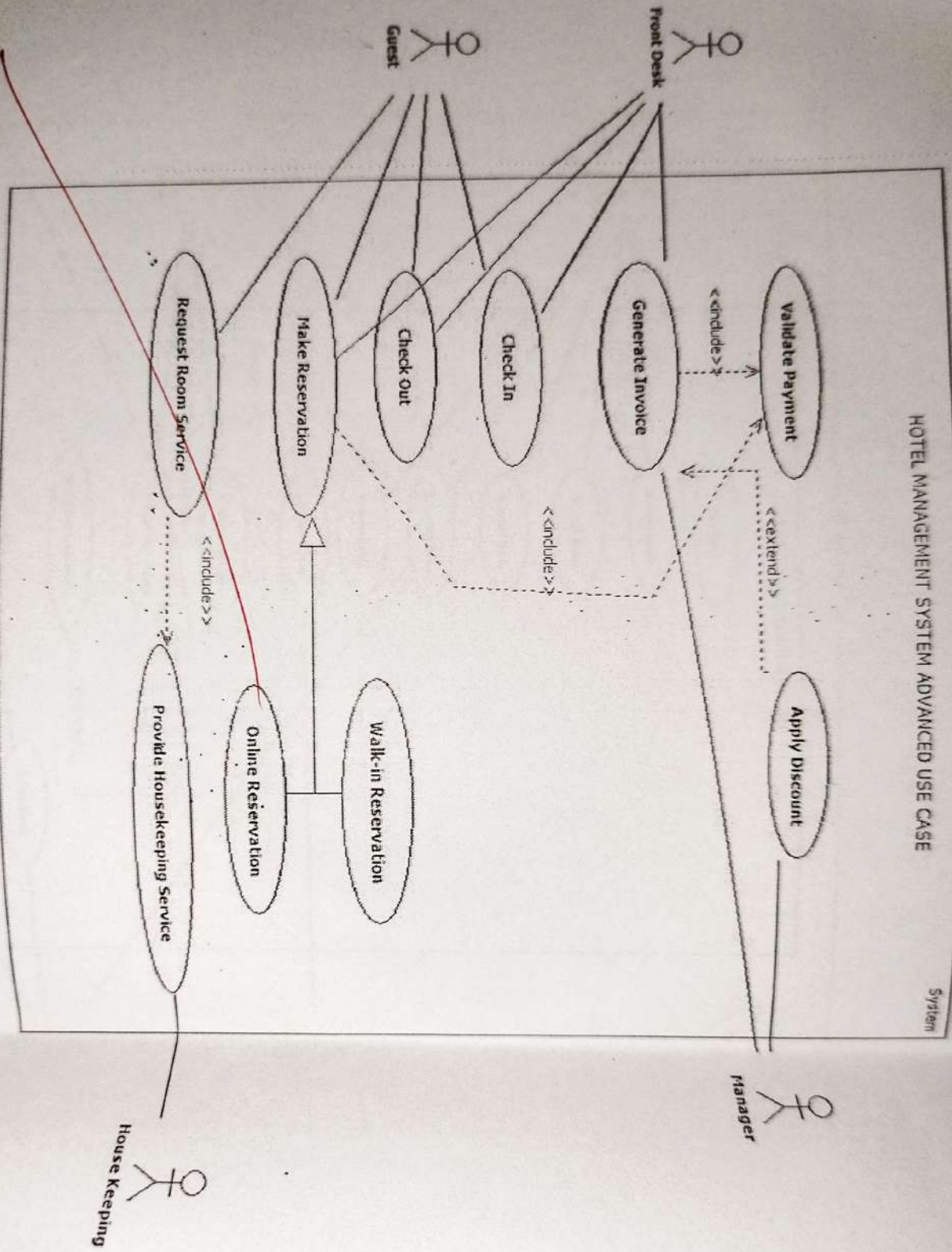




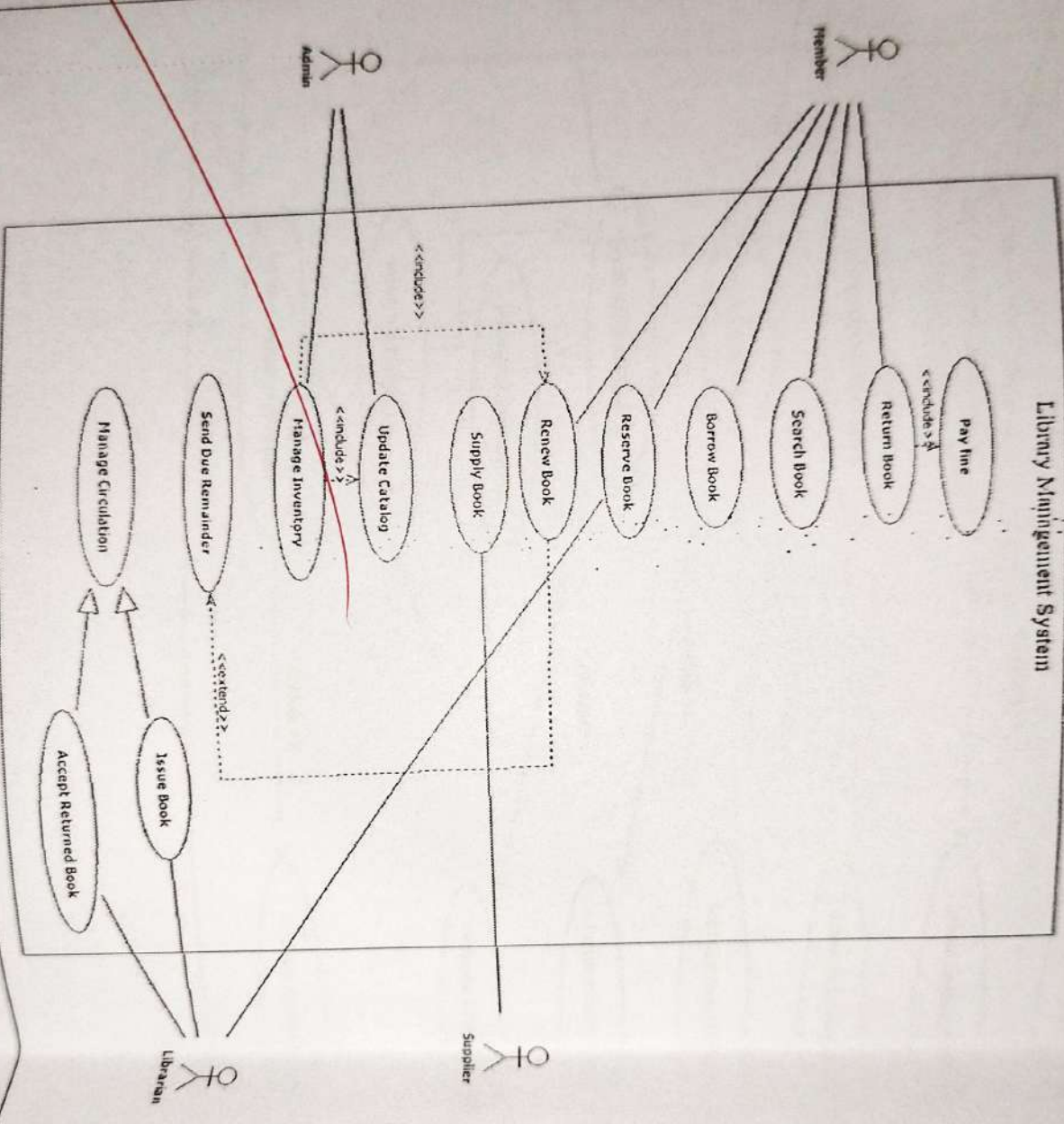


HOTEL MANAGEMENT SYSTEM ADVANCED USE CASE

System



Library Management System



PASSPORT AUTOMATION SYSTEM

